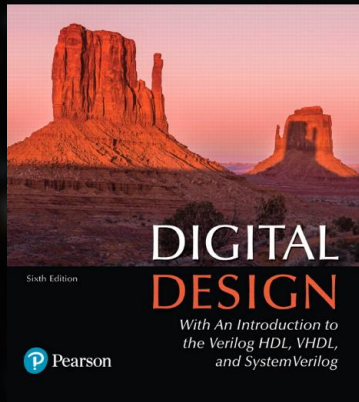




LEC03 & LAB03

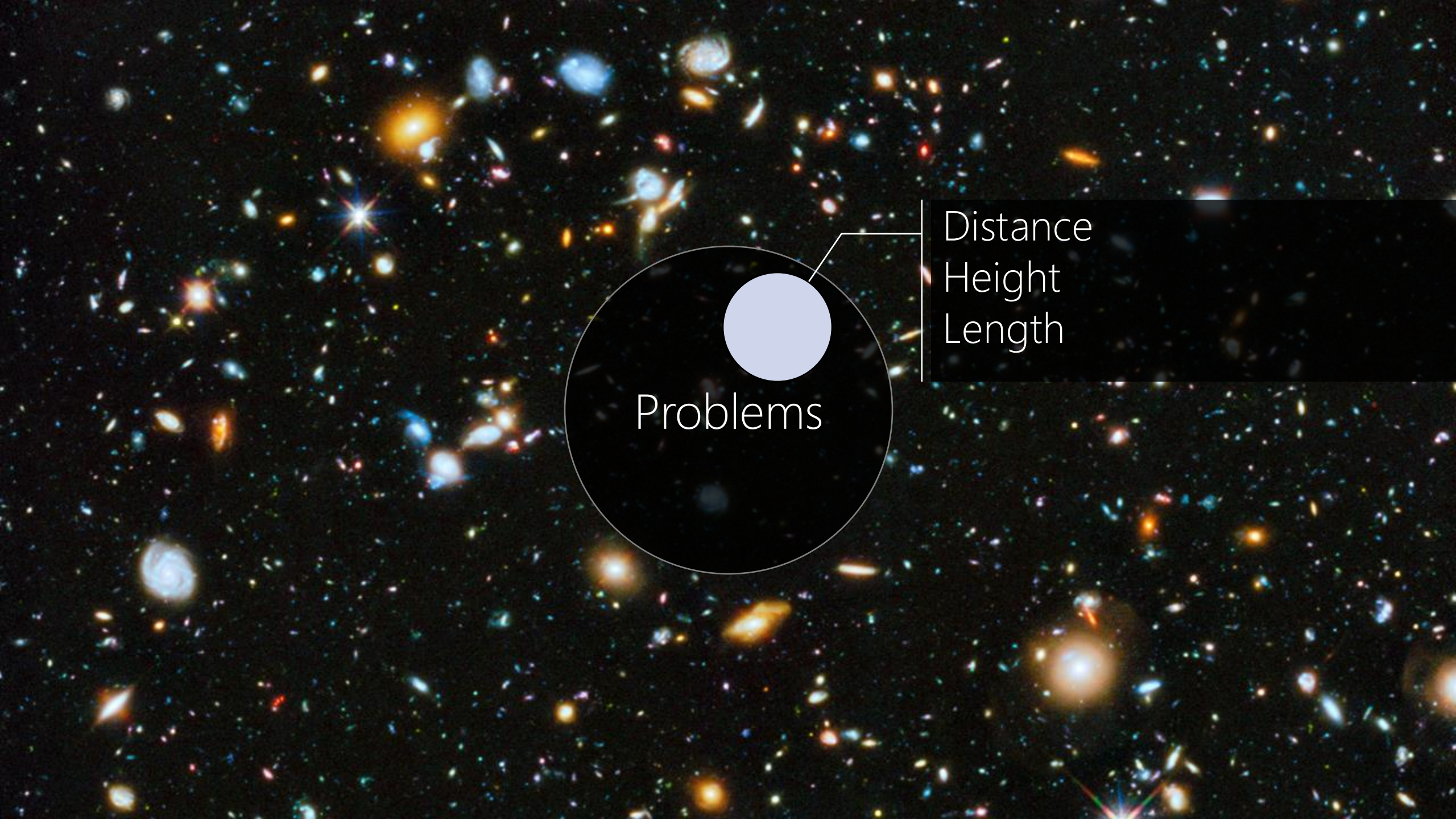
Week03 >> Lec03: Signed Numbers
Week03 >> Lab03: Signed Complement

M. Morris Mano • Michael D. Ciletti



Chapter 1

Digital Systems and Binary Numbers

The background is a deep-field astronomical image showing a vast number of galaxies in various colors (blue, orange, red, white) against a black space. In the center, there is a diagram consisting of a large black circle with a smaller light blue circle inside it. A line connects the light blue circle to a text box on the right. The word "Problems" is written in white inside the large black circle.

Problems

Distance
Height
Length



Quantization

A deep-field astronomical image showing a vast field of galaxies and stars against a black background. The galaxies are in various colors (blue, orange, white) and shapes (spiral, elliptical, irregular). The stars are small, bright points of light. Two horizontal blue lines are positioned above and below the title text.

NUMBER SYSTEMS

QUATERNARY SYSTEM

/kwaa·tur·neh·ree/

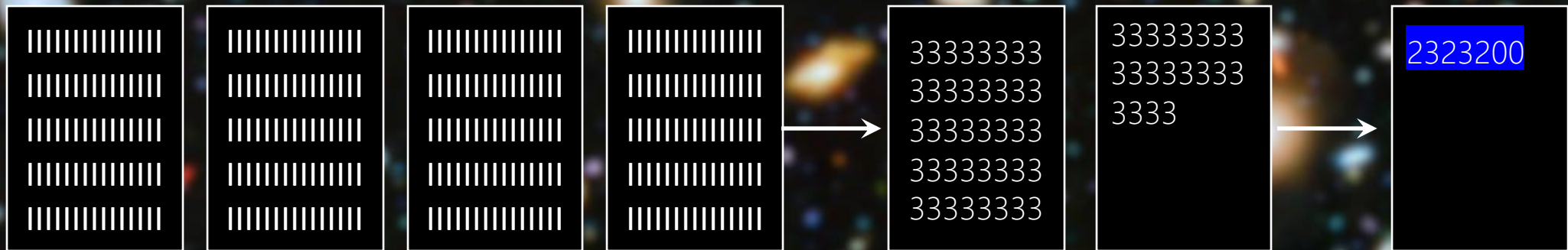
aka. Base-4, Radix-4

$(0,1,2,3)_4$

Hindu-Arabic Numerals
Originated in India
7th Century AD

$$N = 12,000 \rightarrow (2323200)_4$$

We'll see how to convert from decimal to base-4 or any other number systems later. Stay tuned!



The image features a cosmic background filled with numerous galaxies of various colors (blue, orange, white) and shapes, set against a dark space. Two horizontal blue lines are positioned above and below the central text. The text "COMMON NUMBER SYSTEMS" is written in a white, sans-serif, all-caps font, centered horizontally.

COMMON NUMBER SYSTEMS



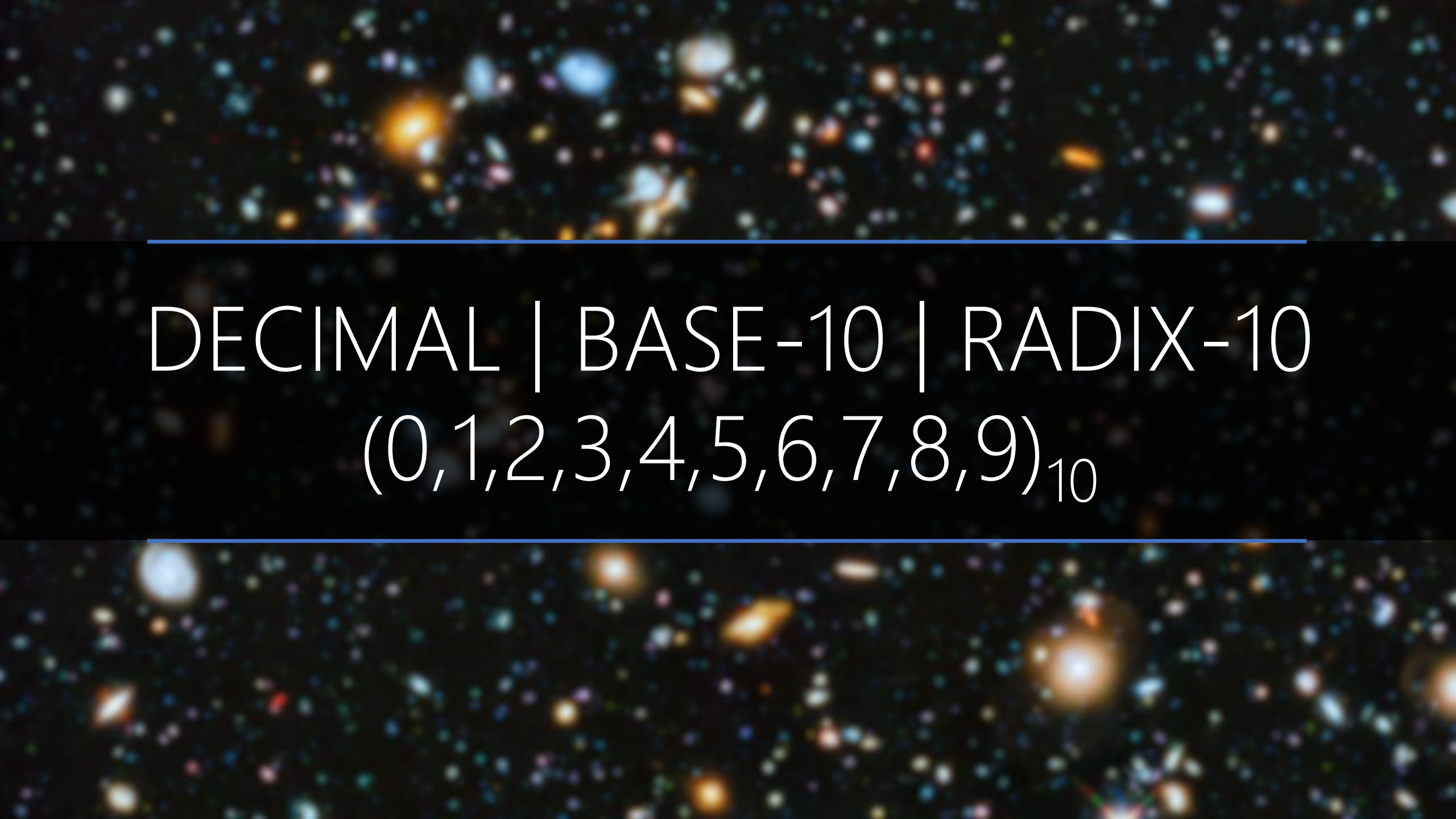
A cosmic background image featuring a dense field of galaxies and stars. The galaxies are in various stages of evolution, with some appearing as bright, irregular shapes and others as more structured, spiral or elliptical forms. The colors range from deep blues and purples to bright oranges and yellows, set against a dark, star-filled sky.

BINARY | BASE-2 | RADIX-2

$(0,1)_2$



OCTAL | BASE-8 | RADIX-8
(0,1,2,3,4,5,6,7)₈

A cosmic background image featuring a dense field of galaxies in various colors (blue, orange, white) against a black space. Two horizontal blue lines frame the central text.

DECIMAL | BASE-10 | RADIX-10
(0,1,2,3,4,5,6,7,8,9)₁₀

Arabic

Eastern Arabic

0123456789

٠ ١ ٢ ٣ ٤ ٥ ٦ ٧ ٨ ٩

*"the **hand** makes the two complementary aspects of integers entirely intuitive. It serves as an instrument permitting natural movement between cardinal and ordinal numbering. If you need to show that a set contains three, four, seven or ten elements, you raise or bend **simultaneously** three, four, seven or ten fingers, using your hand as cardinal mapping. If you want to count out the same things, then you bend or raise three, four, seven or ten fingers in **succession**, using the hand as an ordinal counting tool."*

- Georges Ifrah The Universal History of Numbers (Wiley, 2000, pp. 21-22)

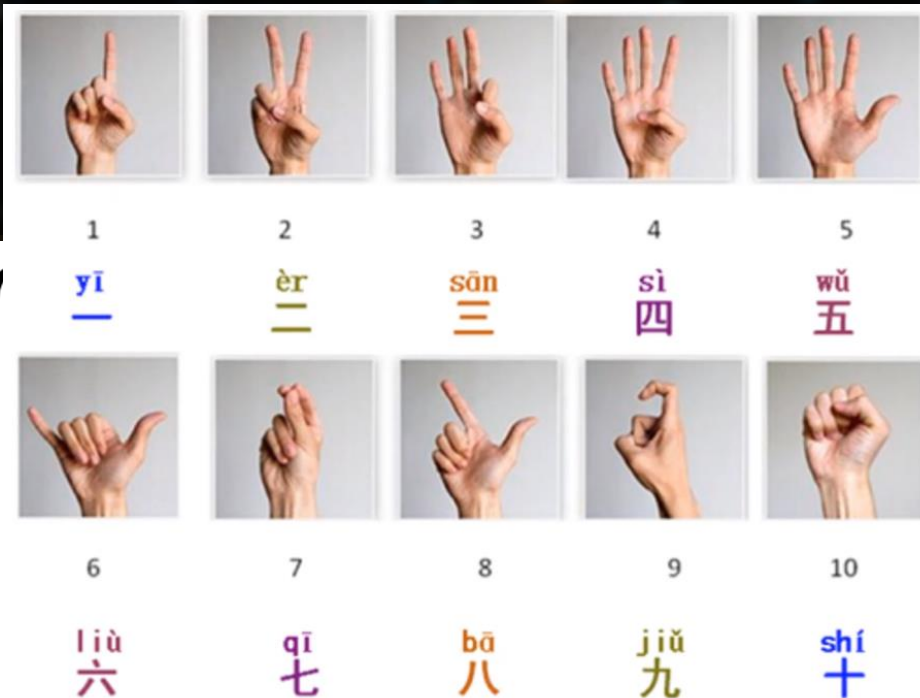
Thanks Cecelia Nydam and Giavi Tran for the hint!

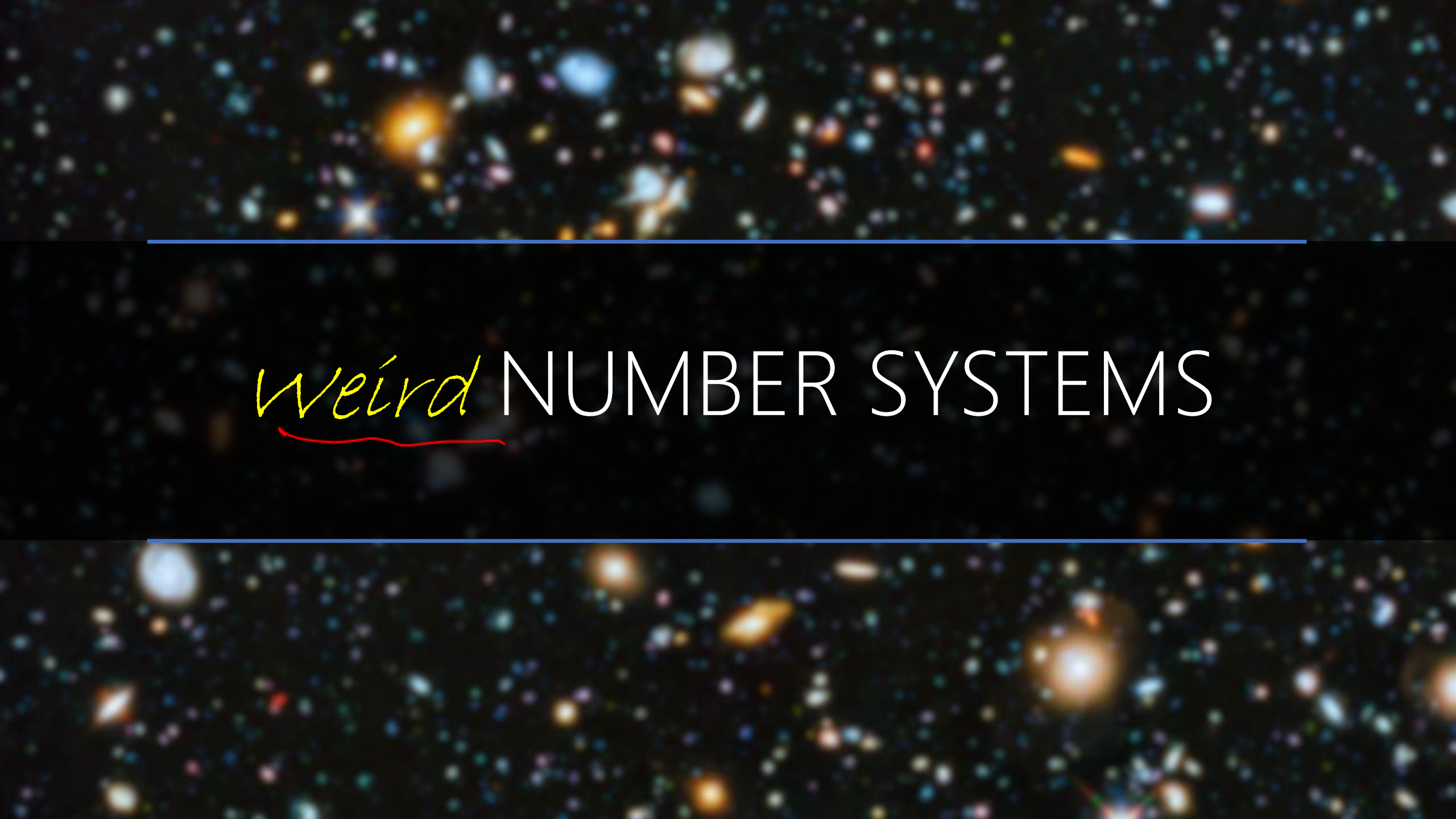
Thai

Chinese

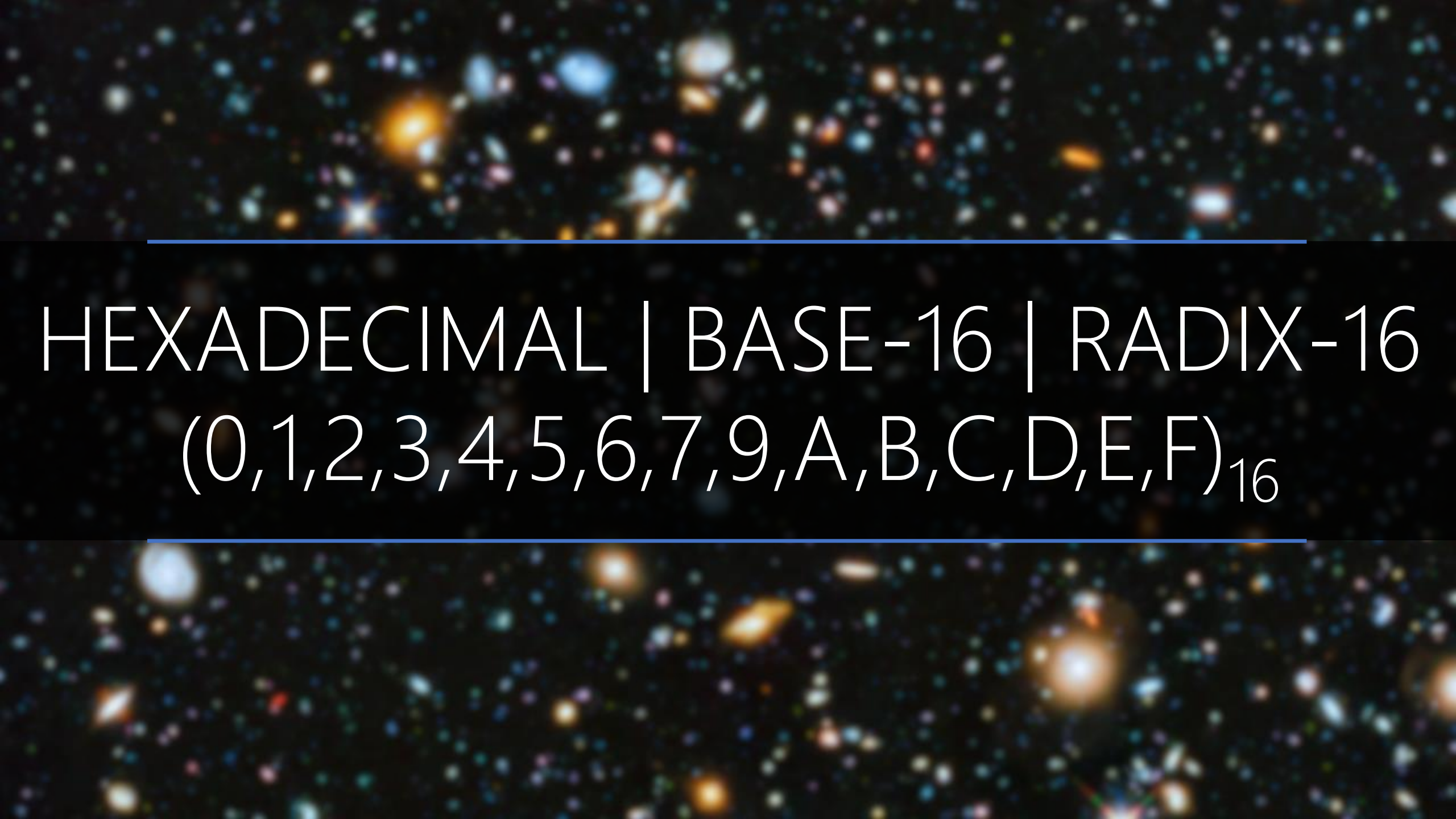
๐ ๑ ๒ ๓ ๔

〇 一 二 三

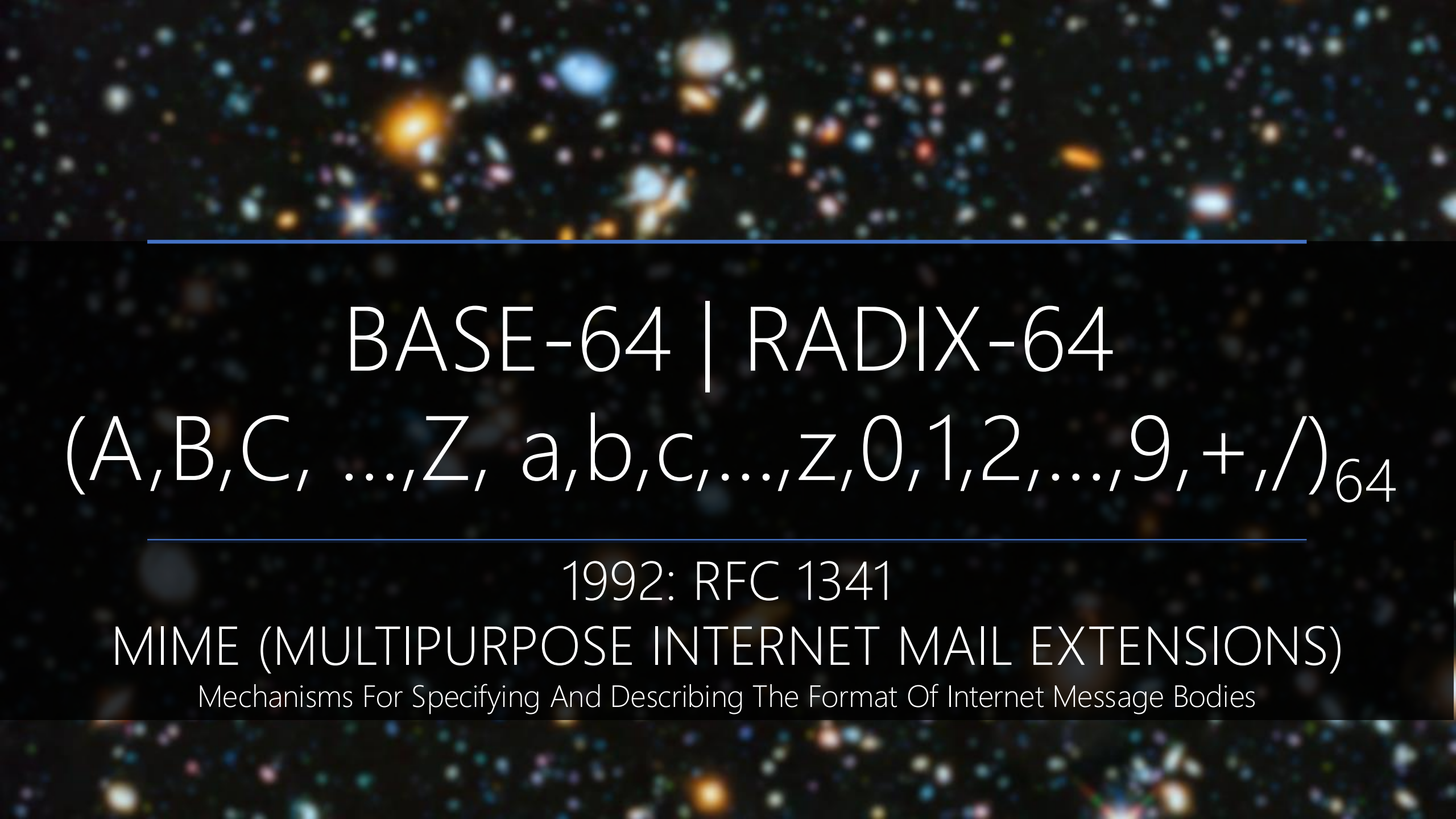


A cosmic background image featuring a dense field of galaxies in various colors (blue, orange, white) against a black space. Two horizontal blue lines are positioned above and below the text.

Weird NUMBER SYSTEMS

A cosmic background image featuring a dense field of galaxies in various colors (blue, orange, white) against a dark space. Two horizontal blue lines frame the central text.

HEXADECIMAL | BASE-16 | RADIX-16
(0,1,2,3,4,5,6,7,9,A,B,C,D,E,F)₁₆



BASE-64 | RADIX-64
(A,B,C, ...,Z, a,b,c,...,z,0,1,2,...,9,+ ,/)₆₄

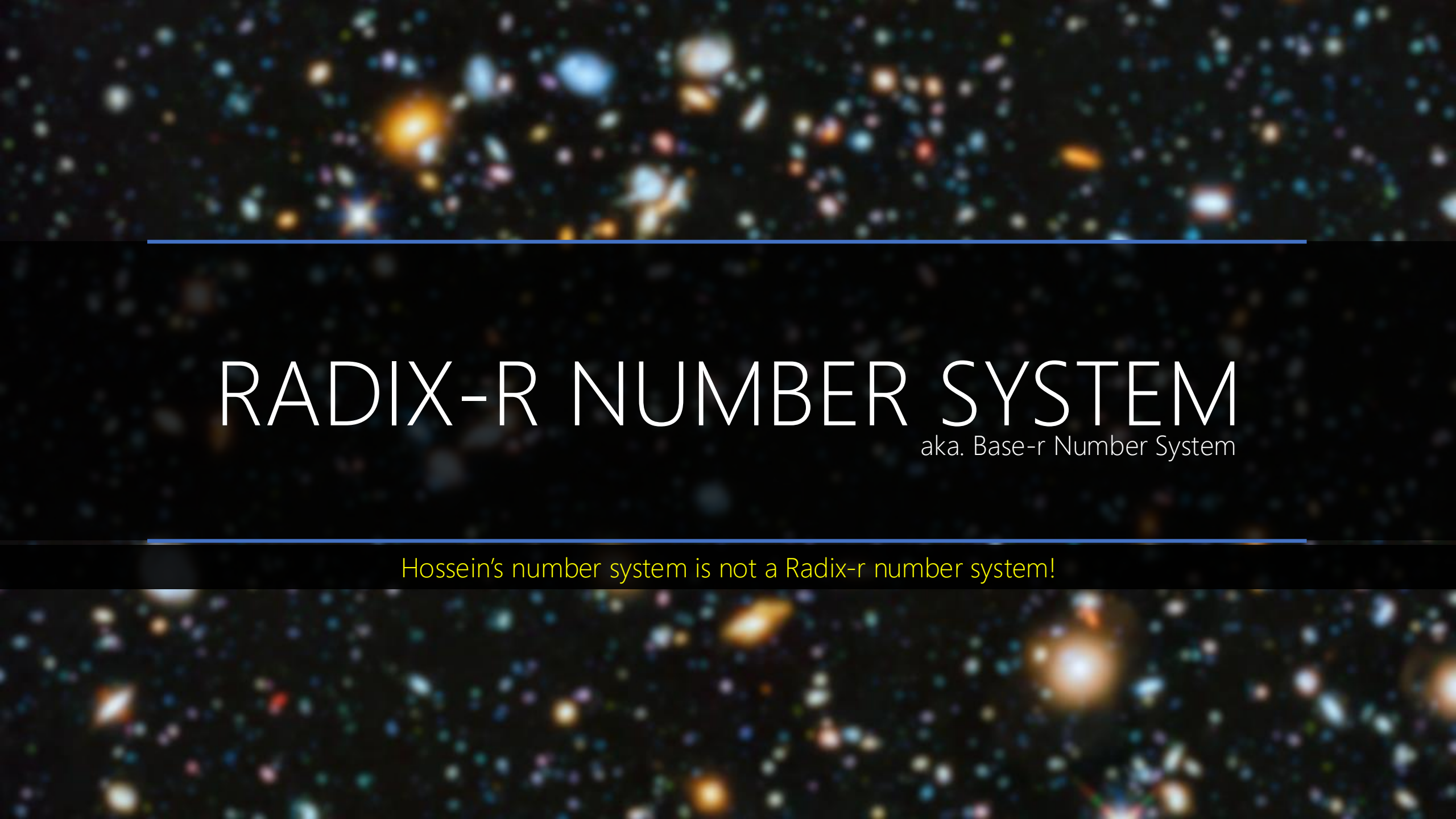
1992: RFC 1341

MIME (MULTIPURPOSE INTERNET MAIL EXTENSIONS)

Mechanisms For Specifying And Describing The Format Of Internet Message Bodies

Digit	Value	→	Digit	Value	→	Digit	Value	→	Digit	Value
A	0		Q	16		g	32		w	48
B	1		R	17		h	33		x	49
C	2		S	18		i	34		y	50
D	3		T	19		j	35		z	51
E	4		U	20		k	36		0	52
F	5		V	21		l	37		1	53
G	6		W	22		m	38		2	54
H	7		X	23		n	39		3	55
I	8		Y	24		o	40		4	56
J	9		Z	25		p	41		5	57
K	10		a	26		q	42		6	58
L	11		b	27		r	43		7	59
M	12		c	28		s	44		8	60
N	13		d	29		t	45		9	61
O	14		e	30		u	46		+	62
P	15		f	31		v	47		/	63

64^7	64^6	64^5	64^4	64^3	64^2	64^1	64^0	
3	a	A	/	d	1	H	+	
55 $\times 64^7$	26 $\times 64^6$	0 $\times 64^5$	63 $\times 64^4$	29 $\times 64^3$	53 $\times 64^2$	7 $\times 64^1$	62 $\times 64^0$	Σ
								243,680,329, 290,238



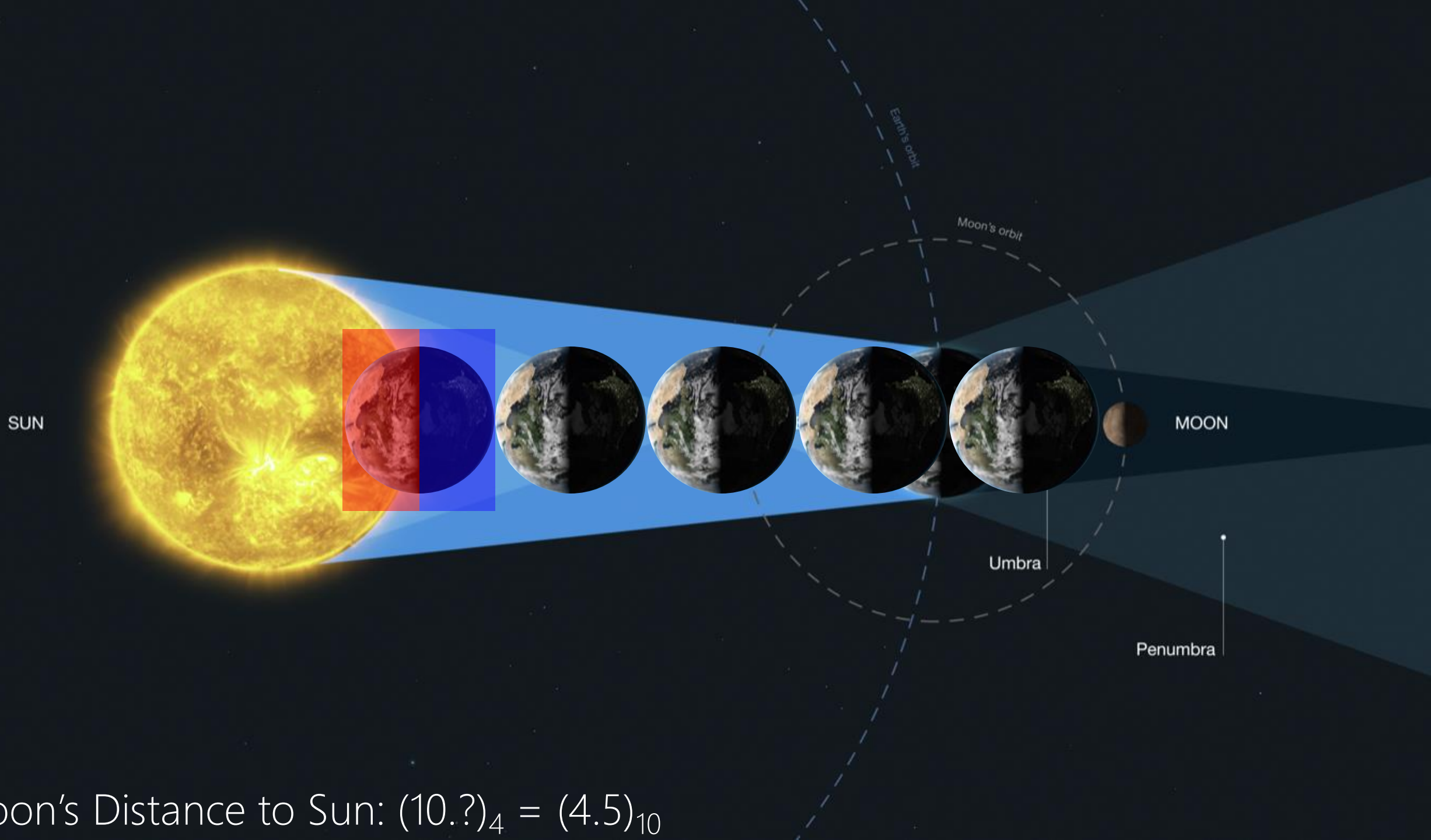
RADIX-R NUMBER SYSTEM

aka. Base-r Number System

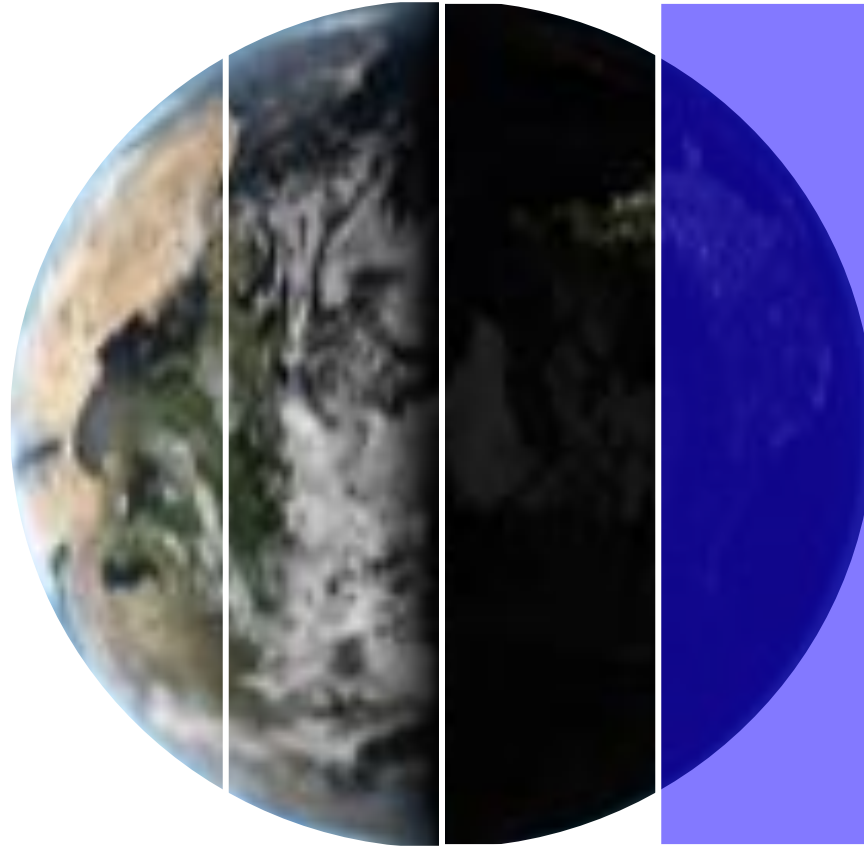
Hossein's number system is not a Radix-r number system!

The image features a deep space background filled with numerous galaxies of various colors (blue, orange, white) and shapes (spiral, elliptical, irregular). Two thin, horizontal blue lines are positioned above and below the central text. The word "FRACTION" is centered in a white, serif font.

FRACTION

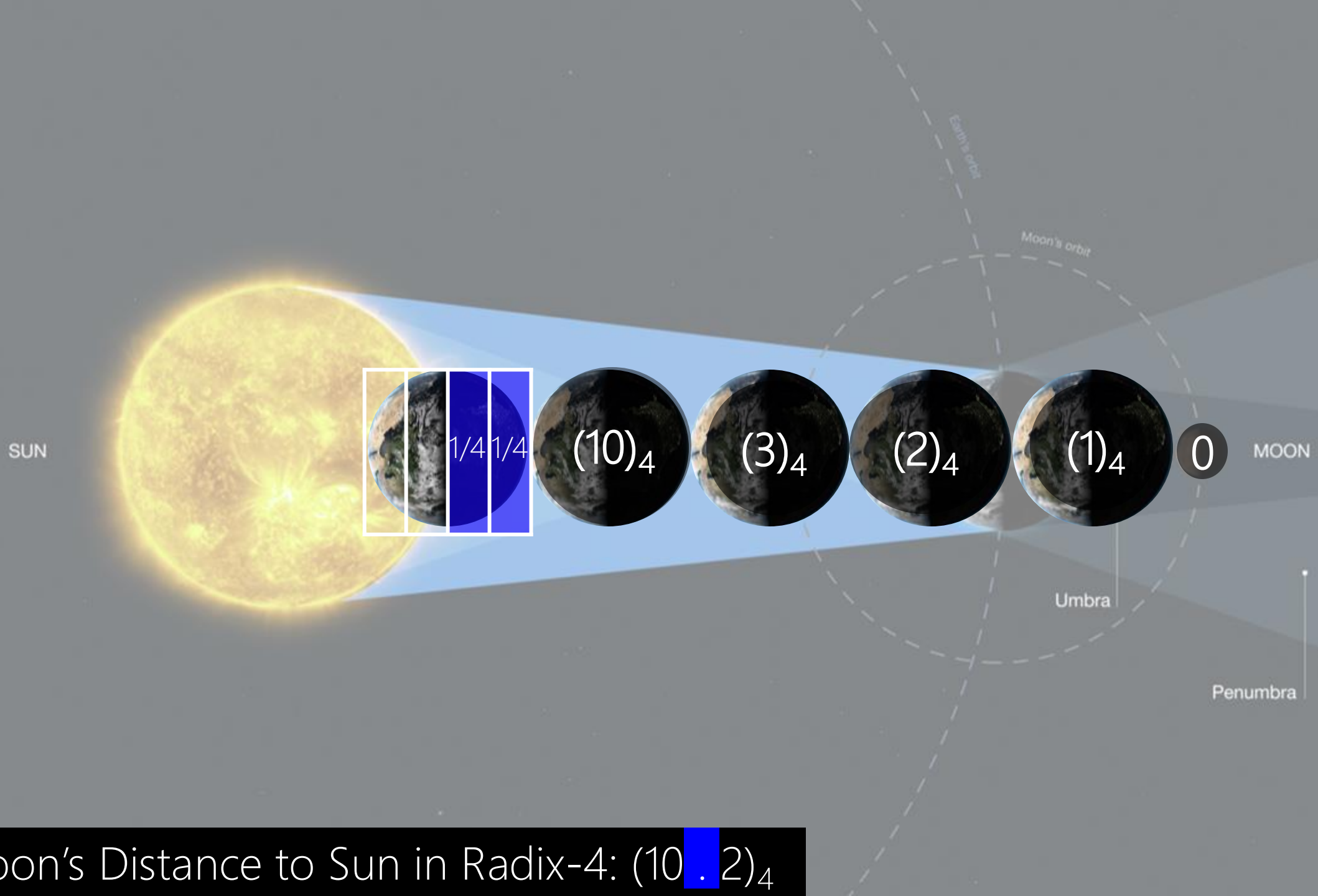


Moon's Distance to Sun: $(10.?)_4 = (4.5)_{10}$

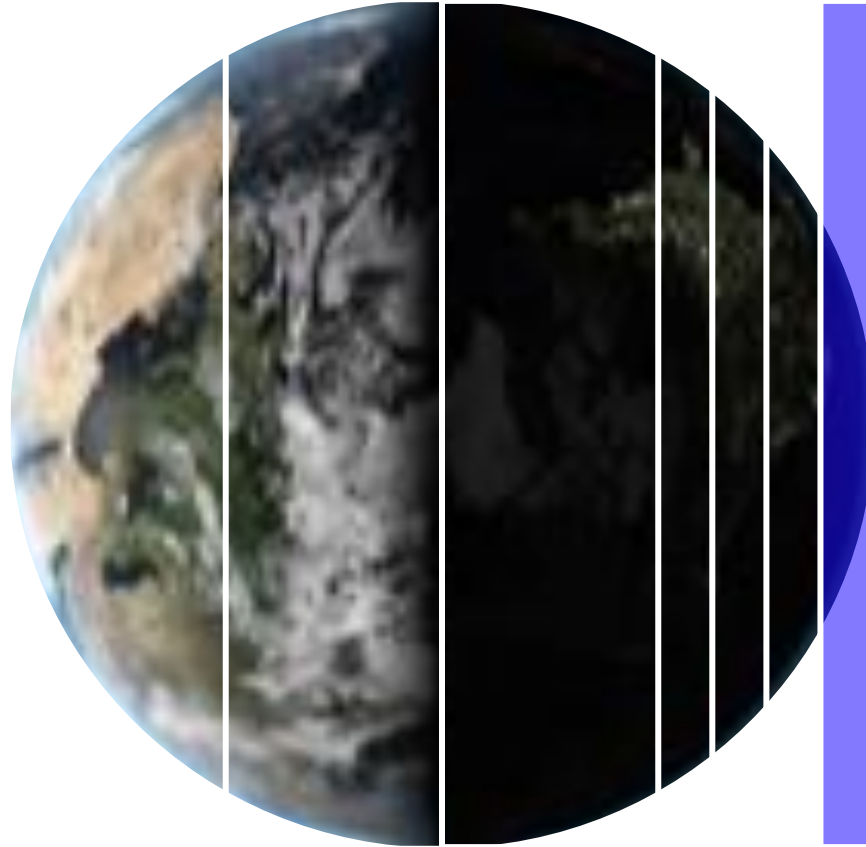


Fraction Point

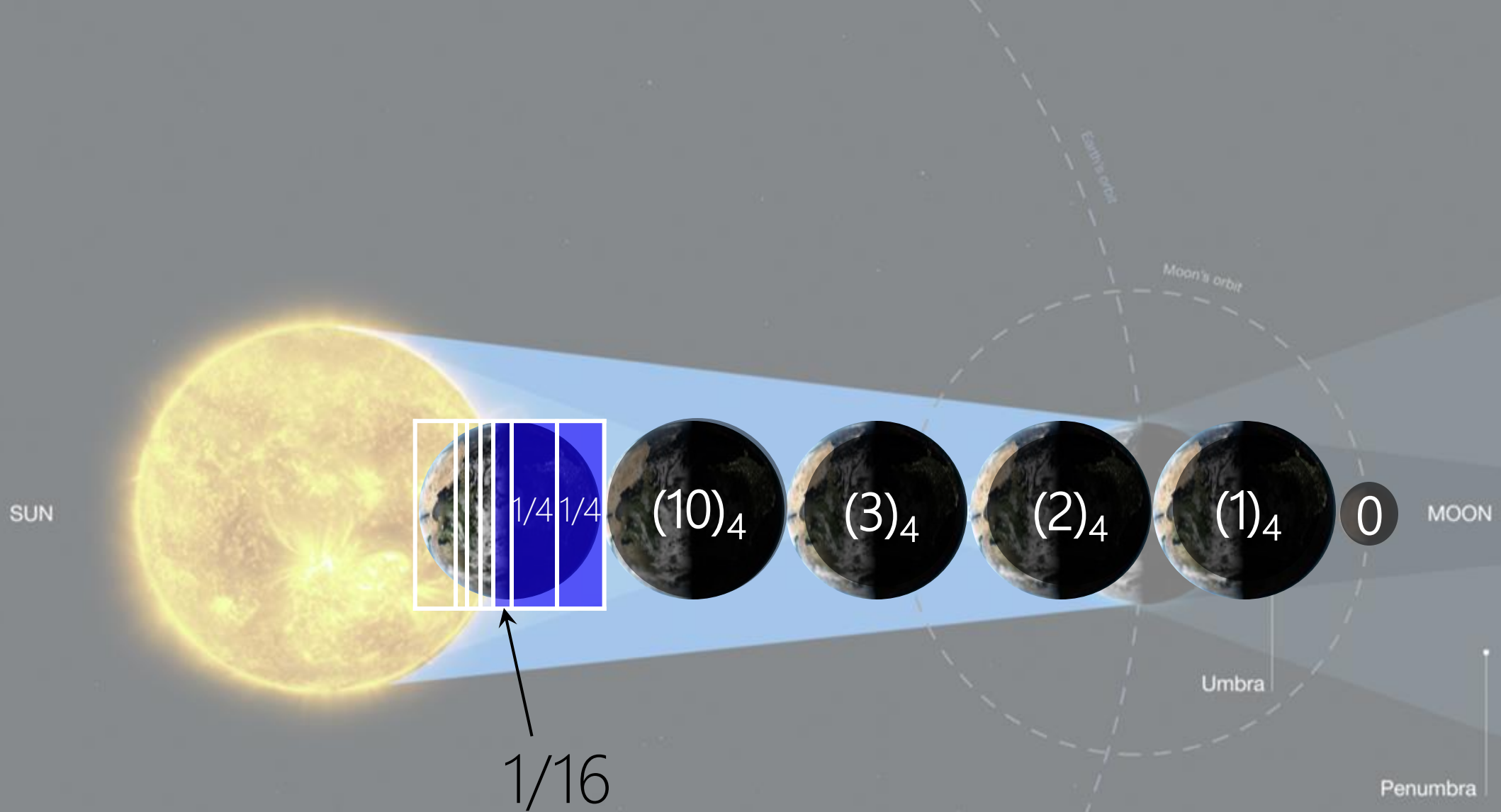
Radix-4 (Base-4) = $1/4$ Earth = 4^{-1} Earth = $(.1)_4$



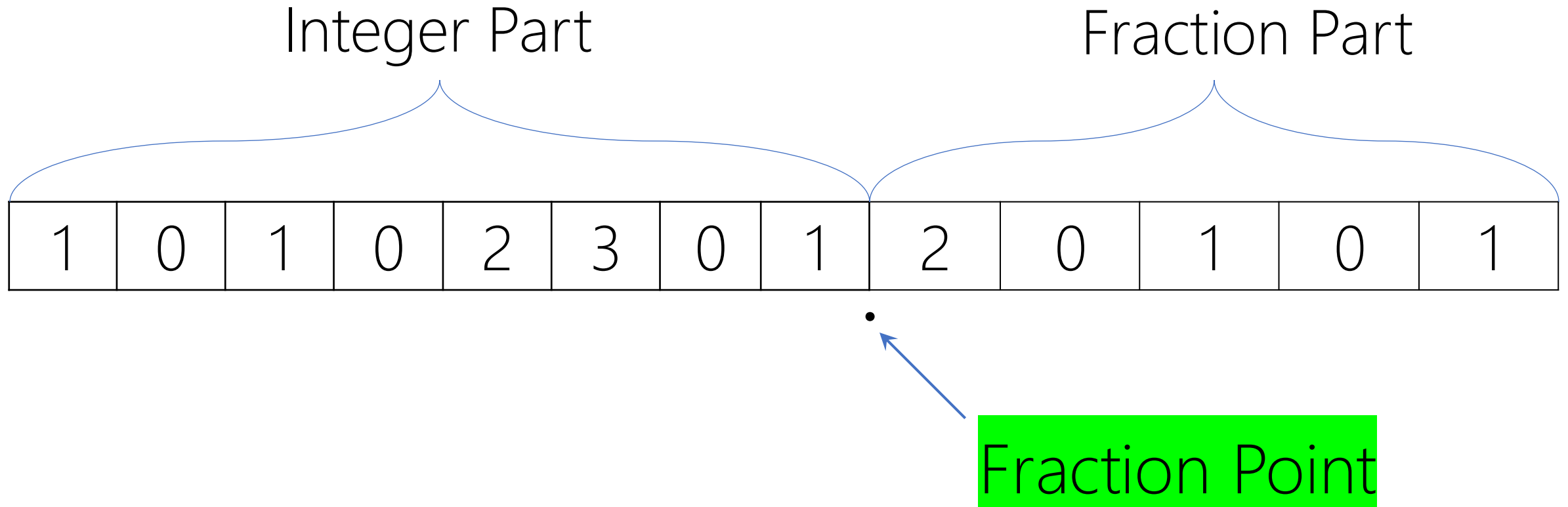
Moon's Distance to Sun in Radix-4: $(10 \cdot 2)_4$



Radix-4 (Base-4) = $(1/4)/4$ Earth = $1/16$ Earth = 4^{-2} Earth



Moon's Distance to Sun in Radix-4: $(10 \text{ . } 21)_4$



We don't waste a position for the fraction point.
It's location is defined already.

Integer Part

4^7	4^6	4^5	4^4	4^3	4^2	4^1	4^0
1	0	1	0	2	3	0	1
1 ×16,384	0	1 ×1,024	0	2 ×64	3 ×16	0	1
				17,584			

×
 .
 Σ
 .

Fraction Part

4^{-1}	4^{-2}	4^{-3}	4^{-4}	4^{-5}
2	0	1	0	1
$\frac{2}{4}$	0	$\frac{1}{64}$	0	$\frac{1}{1,024}$
5166015625				

Let $(N)_r$ be a radix- r (base- r) number in a positional weighting number system, then

$$(N)_r = (d_{n-1}r^{n-1} + \dots + d_0r^0 \cdot d_{-1}r^{-1} + d_{-2}r^{-2} + \dots + d_{-m}r^{-m})_{10}$$

where:

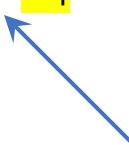
r = radix (base)

d_i = digit at position i , $0 \leq d_i \leq r - 1$

r^i = weight of position i

n = number of digits in integer part of N

m = number of digits in fraction part of N



Fraction Point

Let $(N)_r$ be a radix- r (base- r) number in a positional weighting number system, then

$$\begin{aligned}
 \text{Min} &= (0_{n-1} \cdots 0_1 0_0 \mid 0_{-1} 0_{-2} \cdots 0_{-m-1} 0_{-m})_r = (0.0)_{10} \\
 \text{Max} &= ((r-1)_{n-1} \cdots (r-1)_0 \mid (r-1)_{-1} (r-1)_{-2} \cdots (r-1)_{-m-1} (r-1)_{-m})_r = (r^n - 1 \mid ?)_{10} \\
 \text{Unit} &= (0_{n-1} \cdots 0_1 0_0 \mid 0_{-1} 0_{-2} \cdots 0_{-m-1} 1_{-m})_r = (r^{-m})_{10}
 \end{aligned}$$

where:

r = radix (base)

r^i = weight of position i

n = number of digits in integer part of N

m = number of digits in fraction part of N

Lecture Assignment

A cosmic background image featuring a dense field of galaxies in various colors (blue, orange, white) against a black space. Two horizontal blue lines frame the central text.

PRACTICE RADIX-2

Radix-2									
Integer (n=4)					Fraction (m=3)			Radix-10	
2 ³	2 ²	2 ¹	2 ⁰	.	2 ⁻¹	2 ⁻²	2 ⁻³		
0	0	0	0	.	0	0	0	0	
0	0	0	0	.	0	0	1	1*2 ⁻³ = 1/8 = 0.125	
0	0	0	0	.	0	1	0	1*2 ⁻² + 0*2 ⁻³ = 1/4 = 0.25	
0	0	0	0	.	0	1	1	1*2 ⁻² + 1*2 ⁻³ = 1/4 + 1/8 = 3/8 = 0.375	
0	0	0	0	.	1	0	0	1*2 ⁻¹ + 0*2 ⁻² + 0*2 ⁻³ = ½ = 0.5	
0	0	0	0	.	1	0	1	1*2 ⁻¹ + 0*2 ⁻² + 1*2 ⁻³ = 1/2 + 1/8 = 5/8 = 0.625	
0	0	0	0	.	1	1	0	1*2 ⁻¹ + 1*2 ⁻² + 0*2 ⁻³ = 1/2 + 1/4 = 3/4 = 0.75	
0	0	0	0	.	1	1	1	1*2 ⁻¹ + 1*2 ⁻² + 1*2 ⁻³ = 1/2 + 1/4 + 1/8 = 0.875	
0	0	0	1	.	0	0	0	1*2 ⁰ + 0*2 ⁻¹ + 0*2 ⁻² + 0*2 ⁻³ = 1	
0	0	0	1	.	0	0	1	1.125	
0	0	0	1	.	0	1	0	1.25	
0	0	0	1	.	0	1	1	1.375	
0	0	0	1	.	1	0	0	1.5	
0	0	0	1	.	1	0	1	1.625	
0	0	0	1	.	1	1	0	1.75	
0	0	0	1	.	1	1	1	1.875	
0	0	1	0	.	0	0	0	2	

What is the **max** in this system with these spaces?
 $(1111.111)_2 = (15.875)_{10}$

Radix-2								
Integer (n=4)					Fraction (m=3)			Radix-10
2 ³	2 ²	2 ¹	2 ⁰	.	2 ⁻¹	2 ⁻²	2 ⁻³	
0	0	0	0	.	0	0	0	0
0	0	0	0	.	0	0	1	1*2 ⁻³ = 1/8 = 0.125
0	0	0	0	.	0	1	0	1*2 ⁻² + 0*2 ⁻³ = 1/4 = 0.25
0	0	0	0	.	0	1	1	1*2 ⁻² + 1*2 ⁻³ = 1/4 + 1/8 = 3/8 = 0.375
0	0	0	0	.	1	0	0	1*2 ⁻¹ + 0*2 ⁻² + 0*2 ⁻³ = 1/2 = 0.5
0	0	0	0	.	1	0	1	1*2 ⁻¹ + 0*2 ⁻² + 1*2 ⁻³ = 1/2 + 1/8 = 5/8 = 0.625
0	0	0	0	.	1	1	0	1*2 ⁻¹ + 1*2 ⁻² + 0*2 ⁻³ = 1/2 + 1/4 = 3/4 = 0.75
0	0	0	0	.	1	1	1	1*2 ⁻¹ + 1*2 ⁻² + 1*2 ⁻³ = 1/2 + 1/4 + 1/8 = 0.875
0	0	0	1	.	0	0	0	1*2 ⁰ + 0*2 ⁻¹ + 0*2 ⁻² + 0*2 ⁻³ = 1
0	0	0	1	.	0	0	1	1.125
0	0	0	1	.	0	1	0	1.25
0	0	0	1	.	0	1	1	1.375
0	0	0	1	.	1	0	0	1.5
0	0	0	1	.	1	0	1	1.625
0	0	0	1	.	1	1	0	1.75
0	0	0	1	.	1	1	1	1.875
0	0	1	0	.	0	0	0	2

Is it possible to show the number $(1.02)_{10}$ in this system with these spaces?

No! The numbers in this system increments by 0.125 unit.

Solution?

A. More precision.

A. More fraction positions.

A. More in m! **How much?**

Radix-2									
Integer (n=4)					Fraction (m=3)			Radix-10	
2 ³	2 ²	2 ¹	2 ⁰	.	2 ⁻¹	2 ⁻²	2 ⁻³		
0	0	0	0	.	0	0	0	0	
0	0	0	0	.	0	0	1	1*2 ⁻³ = 1/8 = 0.125	
0	0	0	0	.	0	1	0	1*2 ⁻² + 0*2 ⁻³ = 1/4 = 0.25	
0	0	0	0	.	0	1	1	1*2 ⁻² + 1*2 ⁻³ = 1/4 + 1/8 = 3/8 = 0.375	
0	0	0	0	.	1	0	0	1*2 ⁻¹ + 0*2 ⁻² + 0*2 ⁻³ = ½ = 0.5	
0	0	0	0	.	1	0	1	1*2 ⁻¹ + 0*2 ⁻² + 1*2 ⁻³ = 1/2 + 1/8 = 5/8 = 0.625	
0	0	0	0	.	1	1	0	1*2 ⁻¹ + 1*2 ⁻² + 0*2 ⁻³ = 1/2 + 1/4 = 3/4 = 0.75	
0	0	0	0	.	1	1	1	1*2 ⁻¹ + 1*2 ⁻² + 1*2 ⁻³ = 1/2 + 1/4 + 1/8 = 0.875	
0	0	0	1	.	0	0	0	1*2 ⁰ + 0*2 ⁻¹ + 0*2 ⁻² + 0*2 ⁻³ = 1	
0	0	0	1	.	0	0	1	1.125	
0	0	0	1	.	0	1	0	1.25	
0	0	0	1	.	0	1	1	1.375	
0	0	0	1	.	1	0	0	1.5	
0	0	0	1	.	1	0	1	1.625	
0	0	0	1	.	1	1	0	1.75	
0	0	0	1	.	1	1	1	1.875	
0	0	1	0	.	0	0	0	2	

Is it possible to show the number $(1.02)_{10}$ in this system with these spaces?

No! The numbers in this system increments by 0.125 unit.

Solution?
B. Find the closest number
 $(1.000)_2 = (1)_{10} \Rightarrow \text{Error} = 0.02$
 $(1.001)_2 = (1.125)_{10} \Rightarrow \text{Error} = 0.105$

Radix-2									
Integer (n=4)					Fraction (m=3)			Radix-10	
2 ³	2 ²	2 ¹	2 ⁰	.	2 ⁻¹	2 ⁻²	2 ⁻³		
0	0	0	0	.	0	0	0	0	
0	0	0	0	.	0	0	1	1*2 ⁻³ = 1/8 = 0.125	
0	0	0	0	.	0	1	0	1*2 ⁻² + 0*2 ⁻³ = 1/4 = 0.25	
0	0	0	0	.	0	1	1	1*2 ⁻² + 1*2 ⁻³ = 1/4 + 1/8 = 3/8 = 0.375	
0	0	0	0	.	1	0	0	1*2 ⁻¹ + 0*2 ⁻² + 0*2 ⁻³ = ½ = 0.5	
0	0	0	0	.	1	0	1	1*2 ⁻¹ + 0*2 ⁻² + 1*2 ⁻³ = 1/2 + 1/8 = 5/8 = 0.625	
0	0	0	0	.	1	1	0	1*2 ⁻¹ + 1*2 ⁻² + 0*2 ⁻³ = 1/2 + 1/4 = 3/4 = 0.75	
0	0	0	0	.	1	1	1	1*2 ⁻¹ + 1*2 ⁻² + 1*2 ⁻³ = 1/2 + 1/4 + 1/8 = 0.875	
0	0	0	1	.	0	0	0	1*2 ⁰ + 0*2 ⁻¹ + 0*2 ⁻² + 0*2 ⁻³ = 1	
0	0	0	1	.	0	0	1	1.125	
0	0	0	1	.	0	1	0	1.25	
0	0	0	1	.	0	1	1	1.375	
0	0	0	1	.	1	0	0	1.5	
0	0	0	1	.	1	0	1	1.625	
0	0	0	1	.	1	1	0	1.75	
0	0	0	1	.	1	1	1	1.875	
0	0	1	0	.	0	0	0	2	

Is it possible to show the number $(1.02)_{10}$ in this system with these spaces?

No! The numbers in this system increments by 0.125 unit.

Solution?
B. Find the closest number
 $(1.000)_2 = (1)_{10} \Rightarrow \text{Error} = 0.02$
 $(1.001)_2 = (1.125)_{10} \Rightarrow \text{Error} = 0.105$

A cosmic background image featuring a dense field of galaxies and stars. The galaxies are in various stages of evolution, with some appearing as bright, irregular blobs and others as more structured, elongated shapes. The colors range from deep blues and purples to bright oranges and yellows, set against a dark, star-filled sky.

PRACTICE RADIX-4

Radix-4								Radix-10
Integer (n=4)				Fraction (m=3)				
4 ³	4 ²	4 ¹	4 ⁰	.	4 ⁻¹	4 ⁻²	4 ⁻³	
0	0	0	0	.	0	0	0	0
0	0	0	0	.	0	0	1	1*4 ⁻³ = 1/64 = 0.015625
0	0	0	0	.	0	0	2	2*4 ⁻³ = 2/64 = 0.03125
0	0	0	0	.	0	0	3	3*4 ⁻³ = 3/64 = 0.046875
0	0	0	0	.	0	1	0	1*4 ⁻² + 0*4 ⁻² = 1/16 = 0.0625
0	0	0	0	.	0	1	1	1*4 ⁻² + 1*4 ⁻² = 1/16 + 1/64 = 0.078125
0	0	0	0	.	0	1	2	1*4 ⁻² + 2*4 ⁻² = 1/16 + 2/64 = 0.09375
0	0	0	0	.	0	1	3	1*4 ⁻² + 3*4 ⁻² = 1/16 + 3/64 = 0.109375
0	0	0	0	.	0	2	0	2*4 ⁻² + 0*4 ⁻² = 2/16 = 0.125
...								
0	0	0	0	.	3	3	3	3*4 ⁻¹ + 3*4 ⁻² + 3*4 ⁻³ = 0.984375
0	0	0	1	.	0	0	0	1
...								
3	3	3	3	.	3	3	0	3*4 ³ + 3*4 ² + 3*4 ¹ +3*4 ⁰ +3*4 ⁻¹ + 3*4 ⁻² + 0*4 ⁻³ = ?
3	3	3	3	.	3	3	1	3*4 ³ + 3*4 ² + 3*4 ¹ +3*4 ⁰ +3*4 ⁻¹ + 3*4 ⁻² + 1*4 ⁻³ = ?
3	3	3	3	.	3	3	2	3*4 ³ + 3*4 ² + 3*4 ¹ +3*4 ⁰ +3*4 ⁻¹ + 3*4 ⁻² + 2*4 ⁻³ = ?
3	3	3	3	.	3	3	3	255.984375

Is it possible to show the number $(1.02)_{10}$ in this system with these spaces?

No! Why?

- Solution:
- A. More in m
 - B. Find the closest number

Radix-4								Radix-10
Integer (n=4)				Fraction (m=3)				
4 ³	4 ²	4 ¹	4 ⁰	.	4 ⁻¹	4 ⁻²	4 ⁻³	
0	0	0	0	.	0	0	0	0
0	0	0	0	.	0	0	1	1*4 ⁻³ = 1/64 = 0.015625
0	0	0	0	.	0	0	2	2*4 ⁻³ = 2/64 = 0.03125
0	0	0	0	.	0	0	3	3*4 ⁻³ = 3/64 = 0.046875
0	0	0	0	.	0	1	0	1*4 ⁻² + 0*4 ⁻² = 1/16 = 0.0625
0	0	0	0	.	0	1	1	1*4 ⁻² + 1*4 ⁻² = 1/16 + 1/64 = 0.078125
0	0	0	0	.	0	1	2	1*4 ⁻² + 2*4 ⁻² = 1/16 + 2/64 = 0.09375
0	0	0	0	.	0	1	3	1*4 ⁻² + 3*4 ⁻² = 1/16 + 3/64 = 0.109375
0	0	0	0	.	0	2	0	2*4 ⁻² + 0*4 ⁻² = 2/16 = 0.125
...								
0	0	0	0	.	3	3	3	3*4 ⁻¹ + 3*4 ⁻² + 3*4 ⁻³ = 0.984375
0	0	0	1	.	0	0	0	1
0	0	0	1	.	0	0	1	1.015625
0	0	0	1	.	0	0	2	1.03125
...								
3	3	3	3	.	3	3	2	3*4 ³ + 3*4 ² + 3*4 ¹ +3*4 ⁰ +3*4 ⁻¹ + 3*4 ⁻² + 2*4 ⁻³ = ?
3	3	3	3	.	3	3	3	255.984375

Is it possible to show the number $(1.02)_{10}$ in this system with these spaces?

No! Why?

Solution:

A. More in m

B. Find the closest number

$(1.001)_4 = (1.015625)_{10} \Rightarrow \text{Error} = 0.004375$

$(1.002)_4 = (1.03125)_{10} \Rightarrow \text{Error} = 0.01125$

Radix-4								
Integer (n=4)					Fraction (m=3)			Radix-10
4 ³	4 ²	4 ¹	4 ⁰	.	4 ⁻¹	4 ⁻²	4 ⁻³	
0	0	0	0	.	0	0	0	0
0	0	0	0	.	0	0	1	1*4 ⁻³ = 1/64 = 0.015625
0	0	0	0	.	0	0	2	2*4 ⁻³ = 2/64 = 0.03125
0	0	0	0	.	0	0	3	3*4 ⁻³ = 3/64 = 0.046875
0	0	0	0	.	0	1	0	1*4 ⁻² + 0*4 ⁻² = 1/16 = 0.0625
0	0	0	0	.	0	1	1	1*4 ⁻² + 1*4 ⁻² = 1/16 + 1/64 = 0.078125
0	0	0	0	.	0	1	2	1*4 ⁻² + 2*4 ⁻² = 1/16 + 2/64 = 0.09375
0	0	0	0	.	0	1	3	1*4 ⁻² + 3*4 ⁻² = 1/16 + 3/64 = 0.109375
0	0	0	0	.	0	2	0	2*4 ⁻² + 0*4 ⁻² = 2/16 = 0.125
...								
0	0	0	0	.	3	3	3	3*4 ⁻¹ + 3*4 ⁻² + 3*4 ⁻³ = 0.984375
0	0	0	1	.	0	0	0	1
0	0	0	1	.	0	0	1	1.015625
0	0	0	1	.	0	0	2	1.03125
...								
3	3	3	3	.	3	3	2	3*4 ³ + 3*4 ² + 3*4 ¹ +3*4 ⁰ +3*4 ⁻¹ + 3*4 ⁻² + 2*4 ⁻³ = ?
3	3	3	3	.	3	3	3	255.984375

Is it possible to show the number $(1.02)_{10}$ in this system with these spaces?

No! Why?

Solution:

A. More in m

B. Find the closest number

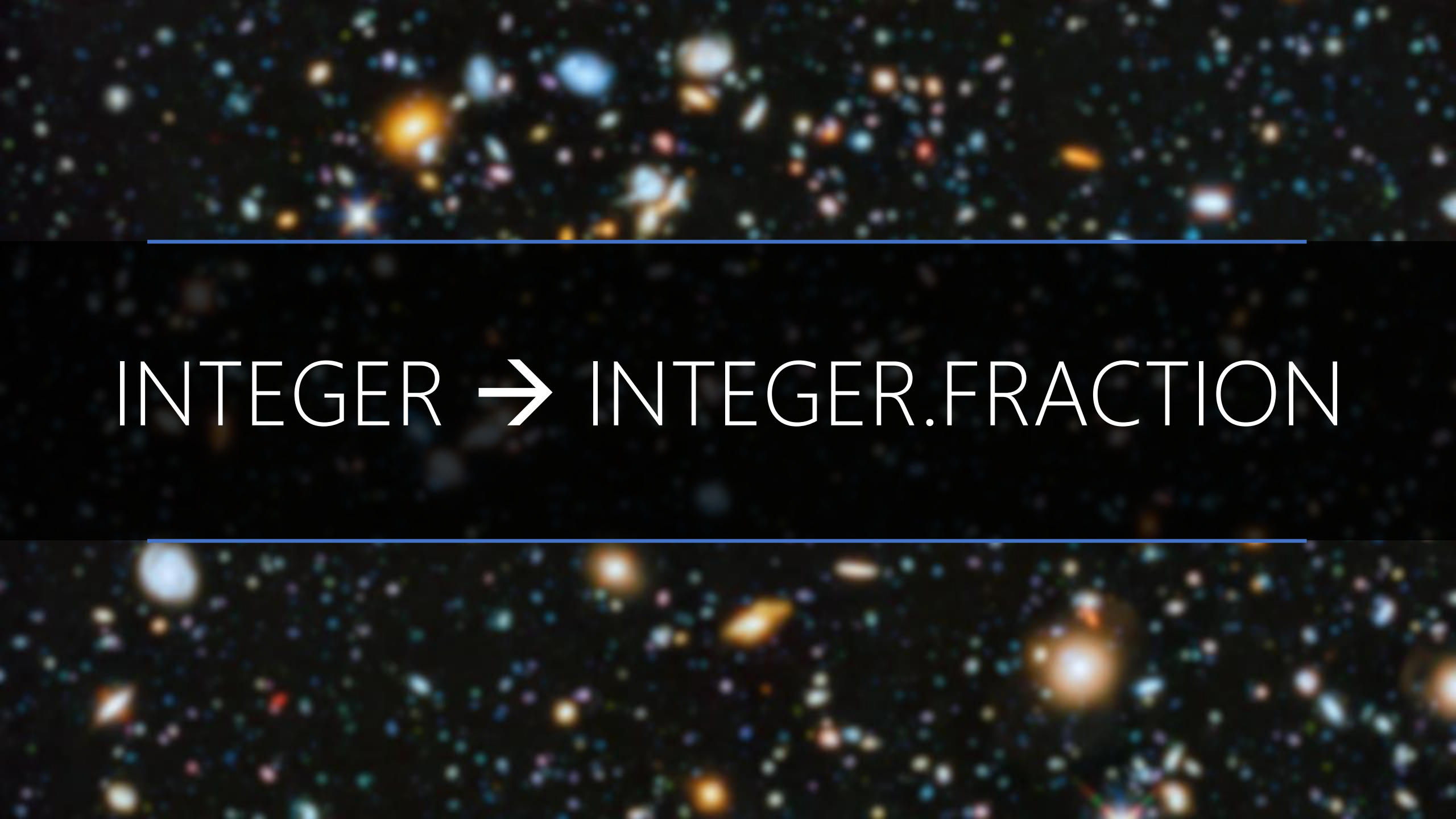
$(1.001)_4 = (1.015625)_{10} \Rightarrow \text{Error} = 0.004375$

$(1.002)_4 = (1.03125)_{10} \Rightarrow \text{Error} = 0.01125$

A cosmic background image featuring a dense field of galaxies and stars. The galaxies are in various stages of evolution, with some appearing as bright, irregular shapes and others as more structured, spiral or elliptical forms. The colors range from deep blues and purples to bright oranges and yellows, set against a dark, star-filled sky.

PRACTICE RADIX-[8,10,16]

At Home

A deep-field astronomical image showing a vast field of galaxies in various colors (blue, orange, white) against a black background. Two horizontal blue lines frame the central text.

INTEGER → INTEGER.FRACTION

The background of the slide is a deep space image showing a vast field of galaxies and stars. The galaxies are in various shapes and sizes, some appearing as bright, fuzzy blobs and others as more distinct, elongated structures. The colors range from bright yellow and orange to deep blues and purples, set against a black background. Two thin, horizontal blue lines are positioned above and below the central text.

CONVERSION

From Base-r to Base-r'

A diagram illustrating a transformation. On the left, a point cloud of white squares is enclosed in large white parentheses, with a subscript r to the right. An arrow points to the right, where a point cloud of blue squares is enclosed in large blue parentheses, with a subscript r' to the right.

Diagram illustrating a transformation from a 5x5 grid of white dots to a 5x5 grid of blue dots, with a large arrow pointing from the first to the second.

we already knew that: sum of the powers of r

Let $(N)_r$ be a radix- r (base- r) number in a positional weighting number system, then

$$(N)_r = (d_{n-1} r^{n-1} + \dots + d_0 r^0 . d_{-1} r^{-1} + d_{-2} r^{-2} + \dots + d_{-m} r^{-m})_{10}$$

where:

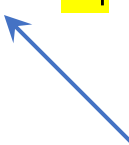
r = radix (base)

d_i = digit at position i , $0 \leq d_i \leq r - 1$

r^i = weight of position i

n = number of digits in integer part of N

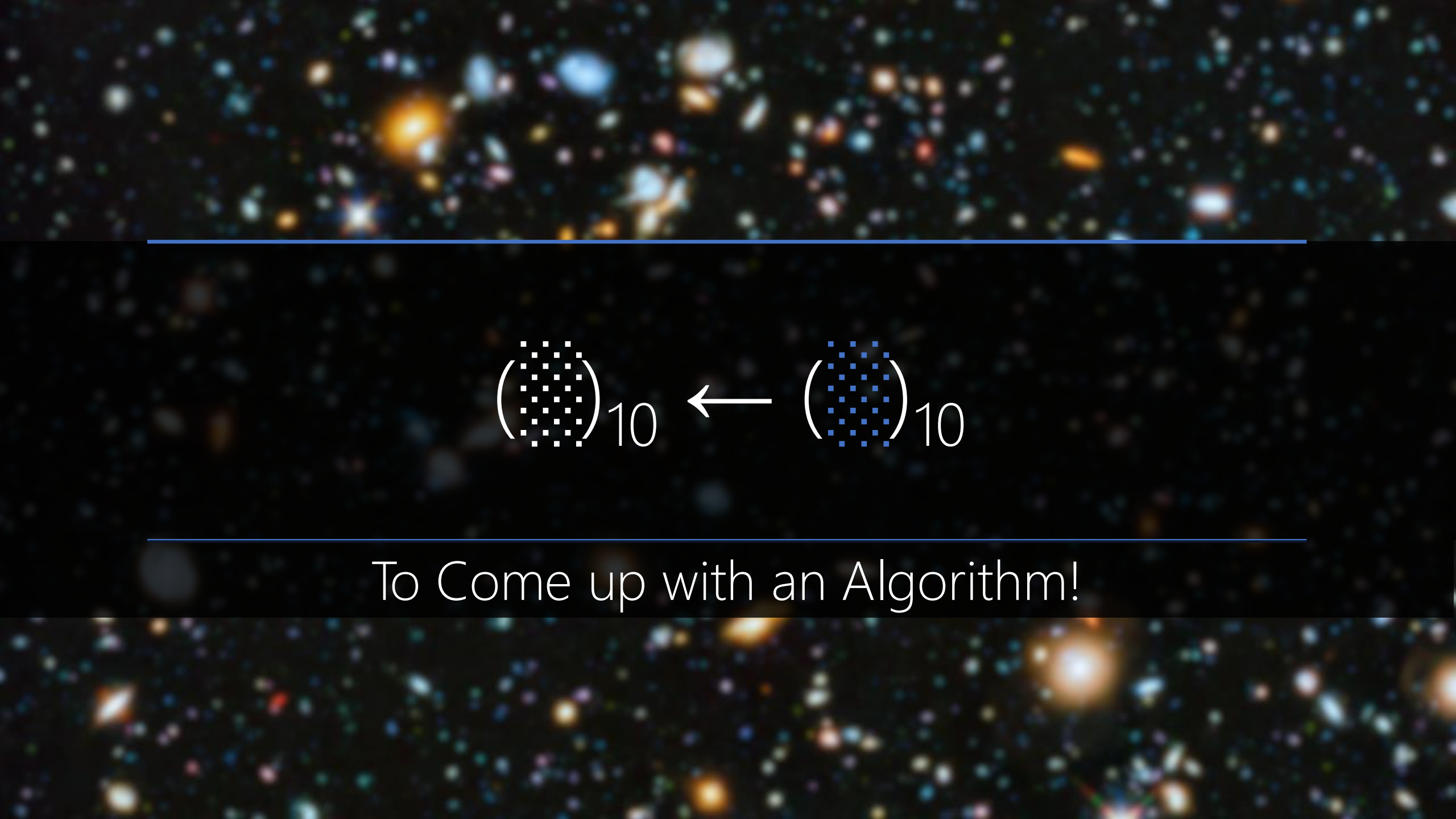
m = number of digits in fraction part of N



Fraction Point

$$\left(\begin{array}{c} \text{10x10 grid of blue squares} \end{array} \right)_{10} \leftarrow \left(\begin{array}{c} \text{10x10 grid of white squares} \end{array} \right)_r$$

Generating the digits of number in base-r



$$\left(\begin{smallmatrix} \bullet & \bullet & \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet & \bullet & \bullet \end{smallmatrix}\right)_{10} \leftarrow \left(\begin{smallmatrix} \bullet & \bullet & \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet & \bullet & \bullet \end{smallmatrix}\right)_{10}$$

To Come up with an Algorithm!

What is the digit in the position with significance of 10^0 ?

(3 0 3 0 2 1 3 1)₁₀

Divide by 10^1 , keep the remainder!

What is the digit in the position with significance of 10^1 ?

(3 0 3 0 2 1 3 1)₁₀

Divide by 10^2 ?

What is the digit in the position with significance of 10^2 ?

(3 0 3 0 2 1 3 1)₁₀

Divide by 10^3 ?

What is the digit in the position with significance of 10^i ?

$$(30302131)_{10} \rightarrow 10^1 \rightarrow 1$$

$$(30302131)_{10} \rightarrow 10^2 \rightarrow 3$$

$$(30302131)_{10} \rightarrow 10^3 \rightarrow 1$$

$$(30302131)_{10} \rightarrow 10^4 \rightarrow 2$$

$$(30302131)_{10} \rightarrow 10^5 \rightarrow 0$$

$$(30302131)_{10} \rightarrow 10^6 \rightarrow 3$$

$$(30302131)_{10} \rightarrow 10^7 \rightarrow 0$$

$$(30302131)_{10} \rightarrow 10^8 \rightarrow 3$$

Divide by 10?

(3 0 3 0 2 1 3 1)₁₀

$$30,302,131 \div 10 = 3,030,213$$

$$30,302,131 \% 10 = 1$$

$$\frac{30,302,131}{10} = 3,030,213 \text{ } r \text{ } 1$$

Divide by 10?

(3 0 3 0 2 1 3 1)₁₀

$$3,030,213 \div 10 = 303,021$$

$$3,030,213 \% 10 = 3$$

$$\frac{3,030,213}{10} = 303,021 \text{ } r \text{ } 3$$

Divide by 10?

(3 0 3 0 2 1 3 1)₁₀

$$303,021 \div 10 = 30,302$$

$$303,021 \% 10 = 1$$

$$\frac{303,021}{10} = 30,302 \text{ } r \text{ } 1$$

What is the digit in the position with significance of 10^i ?

$$(30302131)_{10} \rightarrow 10^1 \rightarrow 1$$

$$(3030213)_{10} \rightarrow 10^1 \rightarrow 3$$

$$(303021)_{10} \rightarrow 10^1 \rightarrow 1$$

$$(30302)_{10} \rightarrow 10^1 \rightarrow 2$$

$$(3030)_{10} \rightarrow 10^1 \rightarrow 0$$

$$(303)_{10} \rightarrow 10^1 \rightarrow 3$$

$$(30)_{10} \rightarrow 10^1 \rightarrow 0$$

$$(3)_{10} \rightarrow 10^1 \rightarrow 3$$

What is the digit in the position with significance of 10^0 ?

$$\frac{30,302,131}{10} = 3,030,213 \text{ } r \text{ } 1$$

Remainder of the 0+1 division by 10!

What is the digit in the position with significance of 10^1 ?

$$\begin{array}{r} 30,302,131 \\ \hline 10 \end{array} = 3,030,213 \text{ } r \text{ } 1$$
$$\begin{array}{r} 3,030,213 \\ \hline 10 \end{array} = 303,021 \text{ } r \text{ } 3$$

Remainder of the 1+1 division by 10!

What is the digit in the position with significance of 10^2 ?

$$\begin{array}{rcl} \frac{30,302,131}{10} & = & 3,030,213 \text{ } r \text{ } 1 \\ & & \frac{30,213}{10} = 3,021,302 \text{ } r \text{ } 3 \\ & & \frac{30,213}{10} = 3,021,302 \text{ } r \text{ } 1 \end{array}$$

Remainder of the $2+1$ division by 10!

What is the digit in the position with significance of 10^3 ?

$$\begin{array}{l} \frac{30,302,131}{10} = 3,030,213 \text{ } r \text{ } 1 \\ \quad \frac{\quad}{10} = 303,021 \text{ } r \text{ } 3 \\ \qquad \frac{\quad}{10} = 30,302 \text{ } r \text{ } 1 \\ \qquad \qquad \frac{\quad}{10} = 3,030 \text{ } r \text{ } 2 \end{array}$$

Remainder of the 3+1 division by 10!

What is the digit in the position with significance of 10^4 ?

$$\begin{array}{rcl} \frac{30,302,131}{10} & = & 3,030,213 \text{ } r \text{ } 1 \\ & \frac{3,030,213}{10} & = 303,021 \text{ } r \text{ } 3 \\ & & \frac{303,021}{10} = 30,302 \text{ } r \text{ } 1 \\ & & & \frac{30,302}{10} = 3,030 \text{ } r \text{ } 2 \\ & & & & \frac{3,030}{10} = 303 \text{ } r \text{ } 0 \end{array}$$

Remainder of the $4+1$ division by $10!$

What is the digit in the position with significance of 10^5 ?

$$\begin{aligned}\frac{30,302,131}{10} &= 3,030,213 \text{ } r \text{ } 1 \\ \frac{3,030,213}{10} &= 303,021 \text{ } r \text{ } 3 \\ \frac{303,021}{10} &= 30,302 \text{ } r \text{ } 1 \\ \frac{30,302}{10} &= 3,030 \text{ } r \text{ } 2 \\ \frac{3,030}{10} &= 303 \text{ } r \text{ } 0 \\ \frac{303}{10} &= 30 \text{ } r \text{ } 3\end{aligned}$$

Remainder of the $5+1$ division by $10!$

What is the digit in the position with significance of 10^6 ?

$$\begin{array}{l} \frac{30,302,131}{10} = 3,030,213 \text{ } r \text{ } 1 \\ \quad \frac{\quad}{10} = 303,021 \text{ } r \text{ } 3 \\ \qquad \frac{\quad}{10} = 30,302 \text{ } r \text{ } 1 \\ \qquad \qquad \frac{\quad}{10} = 3,030 \text{ } r \text{ } 2 \\ \qquad \qquad \qquad \frac{\quad}{10} = 303 \text{ } r \text{ } 0 \\ \qquad \qquad \qquad \qquad \frac{\quad}{10} = 30 \text{ } r \text{ } 3 \\ \qquad \qquad \qquad \qquad \qquad \frac{\quad}{10} = 3 \text{ } r \text{ } 0 \end{array}$$

Remainder of the $6+1$ division by 10!

What is the digit in the position with significance of 10^7 ?

$$\begin{array}{l} \frac{30,302,131}{10} = 3,030,213 \text{ } r \text{ } 1 \\ \frac{\quad \quad \quad}{10} = 303,021 \text{ } r \text{ } 3 \\ \frac{\quad \quad \quad}{10} = 30,302 \text{ } r \text{ } 1 \\ \frac{\quad \quad \quad}{10} = 3,030 \text{ } r \text{ } 2 \\ \frac{\quad \quad \quad}{10} = 303 \text{ } r \text{ } 0 \\ \frac{\quad \quad \quad}{10} = 30 \text{ } r \text{ } 3 \\ \frac{\quad \quad \quad}{10} = 3 \text{ } r \text{ } 0 \\ \frac{\quad \quad \quad}{10} = 0 \text{ } r \text{ } 3 \end{array}$$


Remainder of the $7+1$ division by $10!$

What is the digit in the position with significance of 10^i ?

$$\begin{aligned}
 \frac{30,302,131}{10} &= 3,030,213 \text{ } r \text{ } 1 \\
 \frac{3,030,213}{10} &= 303,021 \text{ } r \text{ } 3 \\
 \frac{303,021}{10} &= 30,302 \text{ } r \text{ } 1 \\
 \frac{30,302}{10} &= 3,030 \text{ } r \text{ } 2 \\
 \frac{3,030}{10} &= 303 \text{ } r \text{ } 0 \\
 \frac{303}{10} &= 30 \text{ } r \text{ } 3 \\
 \frac{30}{10} &= 3 \text{ } r \text{ } 0 \\
 \frac{3}{10} &= 0 \text{ } r \text{ } 3
 \end{aligned}$$

(3 0 3 0 2 1 3 1)₁₀

What is the digit in the position with significance of 16^0 ?

$$\frac{30,302,131}{16} = 1,893,883 \text{ r } 3$$

Remainder of the 0+1 division by 16!

What is the digit in the position with significance of 16^1 ?

$$\frac{30,302,131}{16} = 1,893,883 \text{ r } 3$$
$$\frac{\quad}{16} = 118,367 \text{ r } 11$$

Remainder of the 1+1 division by 16!

What is the digit in the position with significance of 16^2 ?

$$\begin{array}{l} \frac{30,302,131}{16} = 1,893,883 \text{ r } 3 \\ \quad \frac{\quad}{16} = 118,367 \text{ r } 11 \\ \qquad \frac{\quad}{16} = 7,397 \text{ r } 15 \end{array}$$

Remainder of the $2+1$ division by 16!

What is the digit in the position with significance of 16^3 ?

$$\begin{array}{l} \frac{30,302,131}{16} = 1,893,883 \text{ r } 3 \\ \quad \frac{\quad}{16} = 118,367 \text{ r } 11 \\ \qquad \frac{\quad}{16} = 7,397 \text{ r } 15 \\ \qquad \qquad \frac{\quad}{16} = 462 \text{ r } 5 \end{array}$$

Remainder of the 3+1 division by 16!

What is the digit in the position with significance of 16^4 ?

$$\begin{array}{l} \frac{30,302,131}{16} = 1,893,883 \text{ r } 3 \\ \quad \frac{1,893,883}{16} = 118,367 \text{ r } 11 \\ \qquad \frac{118,367}{16} = 7,397 \text{ r } 15 \\ \qquad \qquad \frac{7,397}{16} = 462 \text{ r } 5 \\ \qquad \qquad \qquad \frac{462}{16} = 28 \text{ r } 14 \end{array}$$

Remainder of the $4+1$ division by 16!

What is the digit in the position with significance of 16^5 ?

$$\begin{array}{l} \frac{30,302,131}{16} = 1,893,883 \text{ r } 3 \\ \quad \frac{}{16} = 118,367 \text{ r } 11 \\ \qquad \frac{}{16} = 7,397 \text{ r } 15 \\ \qquad \qquad \frac{}{16} = 462 \text{ r } 5 \\ \qquad \qquad \qquad \frac{}{16} = 28 \text{ r } 14 \\ \qquad \qquad \qquad \qquad \frac{}{16} = 1 \text{ r } 12 \end{array}$$

Remainder of the $5+1$ division by 16!

What is the digit in the position with significance of 16^6 ?

$$\begin{array}{l} \frac{30,302,131}{16} = 1,893,883 \text{ r } 3 \\ \quad \frac{}{16} = 118,367 \text{ r } 11 \\ \qquad \frac{}{16} = 7,397 \text{ r } 15 \\ \qquad \qquad \frac{}{16} = 462 \text{ r } 5 \\ \qquad \qquad \qquad \frac{}{16} = 28 \text{ r } 14 \\ \qquad \qquad \qquad \qquad \frac{}{16} = 1 \text{ r } 12 \\ \qquad \qquad \qquad \qquad \qquad \frac{}{16} = 0 \text{ r } 1 \end{array}$$

Remainder of the $6+1$ division by 16!

What is the digit in the position with significance of 16^6 ?

$$\begin{aligned} \frac{30,302,131}{16} &= 1,893,883 \text{ r } 3 \\ \frac{1,893,883}{16} &= 118,367 \text{ r } 11 \\ \frac{118,367}{16} &= 7,397 \text{ r } 15 \\ \frac{7,397}{16} &= 462 \text{ r } 5 \\ \frac{462}{16} &= 28 \text{ r } 14 \\ \frac{28}{16} &= 1 \text{ r } 12 \\ \frac{1}{16} &= 0 \text{ r } 1 \end{aligned}$$

(1 12 14 5 15 11 3)₁₆

What is the digit in the position with significance of 16^6 ?

$$\begin{aligned}\frac{30,302,131}{16} &= 1,893,883 \text{ r } 1 \\ \frac{1,893,883}{16} &= 118,367 \text{ r } 11 \\ \frac{118,367}{16} &= 7,397 \text{ r } 0 \\ \frac{7,397}{16} &= 462 \text{ r } 5 \\ \frac{462}{16} &= 28 \text{ r } 14 \\ \frac{28}{16} &= 1 \text{ r } 12 \\ \frac{1}{16} &= 0 \text{ r } 1\end{aligned}$$

(1 12 14 5 15 11 3)₁₆

$$1, 2, 3, 4, 5, 6, 7, 8, 9, A = 9 + 1 = (10)_{10}$$

$$B = A + 1 = (11)_{10}$$

$$C = B + 1 = (12)_{10}$$

$$D = C + 1 = (13)_{10}$$

$$E = D + 1 = (14)_{10}$$

$$F = E + 1 = (15)_{10}$$

What is the digit in the position with significance of 16^6 ?

$$\begin{aligned}\frac{30,302,131}{16} &= 1,893,883 \text{ r } 1 \\ \frac{1,893,883}{16} &= 118,367 \text{ r } 11 \\ \frac{118,367}{16} &= 7,397 \text{ r } 0 \\ \frac{7,397}{16} &= 462 \text{ r } 5 \\ \frac{462}{16} &= 28 \text{ r } 14 \\ \frac{28}{16} &= 1 \text{ r } 12 \\ \frac{1}{16} &= 0 \text{ r } 1\end{aligned}$$

(1 C E 5 F B 3)₁₆

What is the digit in the position with significance of 8^i ?

Quotient	Remainder
$30,302,131 \div 8$	3
$3787766 \div 8$	6
$473470 \div 8$	6
$59183 \div 8$	7
$7397 \div 8$	5
$924 \div 8$	4
$115 \div 8$	3
$14 \div 8$	6
$1 \div 8$	1
0	



$(1CE5FB3)_{16}$
 $(30,302,131)_{10}$

 $(163457663)_8$

What is the digit in the position with significance of 4^i ?

Quotient	Remainder
$30,302,131 \div 4$	3
$7575532 \div 4$	0
$1893883 \div 4$	3
$473470 \div 4$	2
$118367 \div 4$	3
$29591 \div 4$	3
$7397 \div 4$	1
$1849 \div 4$	1
$462 \div 4$	2
$115 \div 4$	3
$28 \div 4$	0
$7 \div 4$	3
$1 \div 4$	1
0	



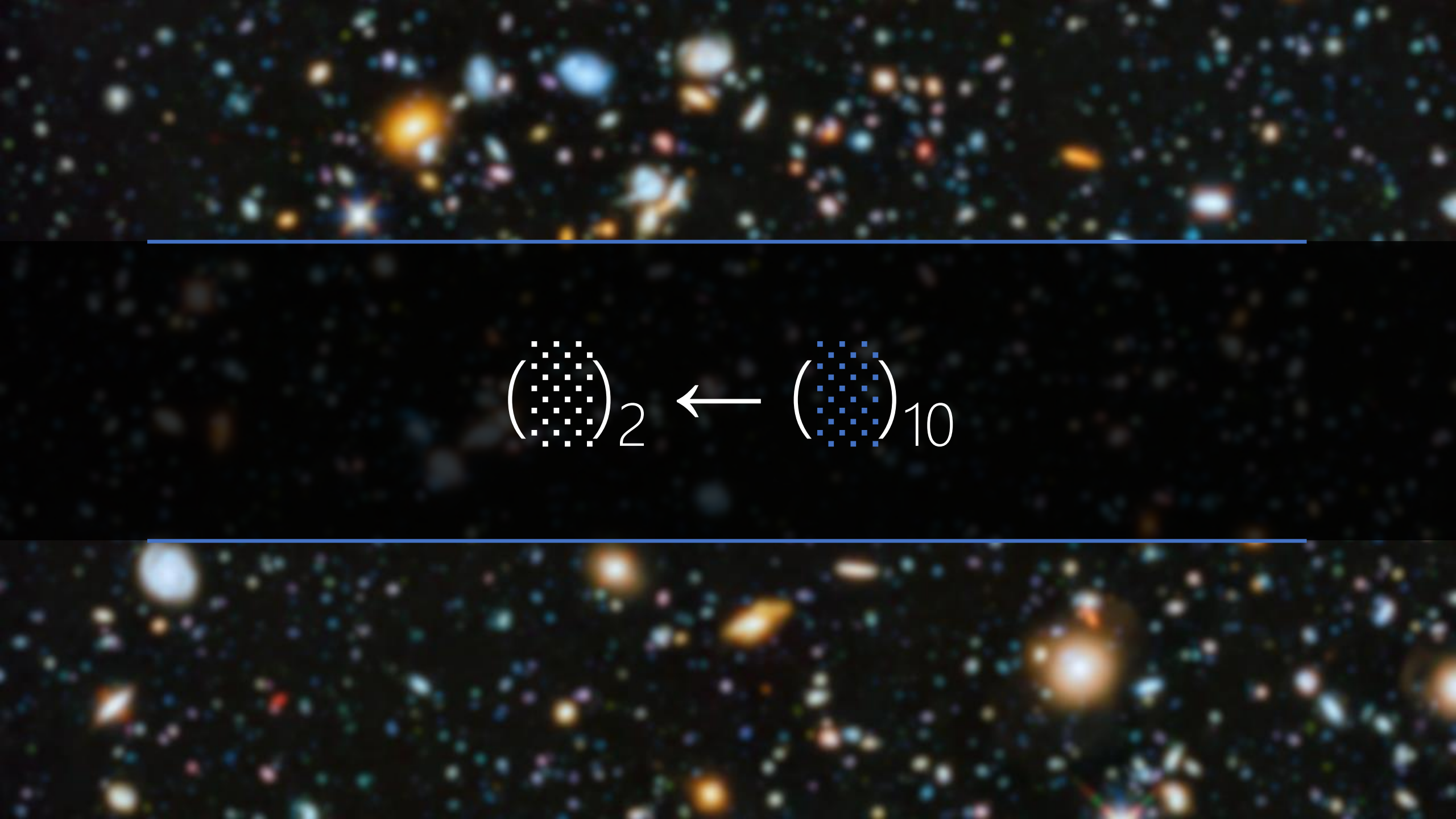
$(1CE5FB3)_{16}$

$(30,302,131)_{10}$

$(163457663)_8$



$(1303211332303)_4$

A deep-field astronomical image showing a vast field of galaxies in various colors (blue, orange, white) against a black background. Two horizontal blue lines are drawn across the image, one above and one below the central text.
$$\left(\begin{smallmatrix} \bullet & \bullet & \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet & \bullet & \bullet \end{smallmatrix}\right)_2 \leftarrow \left(\begin{smallmatrix} \bullet & \bullet & \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet & \bullet & \bullet \end{smallmatrix}\right)_{10}$$

What is the digit in the position with significance of 2^i ?

Quotient	Remainder
30,302,131 ÷ 2	1
15151065 ÷ 2	1
7575532 ÷ 2	0
3787766 ÷ 2	0
1893883 ÷ 2	1
946941 ÷ 2	1
473470 ÷ 2	0
236735 ÷ 2	1
118367 ÷ 2	1
59183 ÷ 2	1
29591 ÷ 2	1
14795 ÷ 2	1
7397 ÷ 2	1
3698 ÷ 2	0
1849 ÷ 2	1
924 ÷ 2	0
462 ÷ 2	0
231 ÷ 2	1
115 ÷ 2	1
57 ÷ 2	1
28 ÷ 2	0
14 ÷ 2	0
7 ÷ 2	1
3 ÷ 2	1
1 ÷ 2	1
0	



$(1CE5FB3)_{16}$

$(30,302,131)_{10}$

$(163457663)_8$

$(1303211332303)_4$

$(111001110010111110110011)_2$

What is the digit in the position with significance of 2^i ?

Quotient	Remainder
30,302,131 ÷ 2	1
15,151,065 ÷ 2	1
7,575,532 ÷ 2	0
3,787,766 ÷ 2	0
1,893,883 ÷ 2	1
946,941 ÷ 2	1
473,470 ÷ 2	0
236,735 ÷ 2	1
118,367 ÷ 2	1
59,183 ÷ 2	1
29,591 ÷ 2	1
14,795 ÷ 2	1
7,397 ÷ 2	1
3,698 ÷ 2	0
1,849 ÷ 2	1
924 ÷ 2	0
462 ÷ 2	0
231 ÷ 2	1
115 ÷ 2	1
57 ÷ 2	1
28 ÷ 2	0
14 ÷ 2	0
7 ÷ 2	1
3 ÷ 2	1
1 ÷ 2	1
0	

(1CE5FB3)₁₆

(30,302,131)₁₀

(163457663)₈

(1303211332303)₄

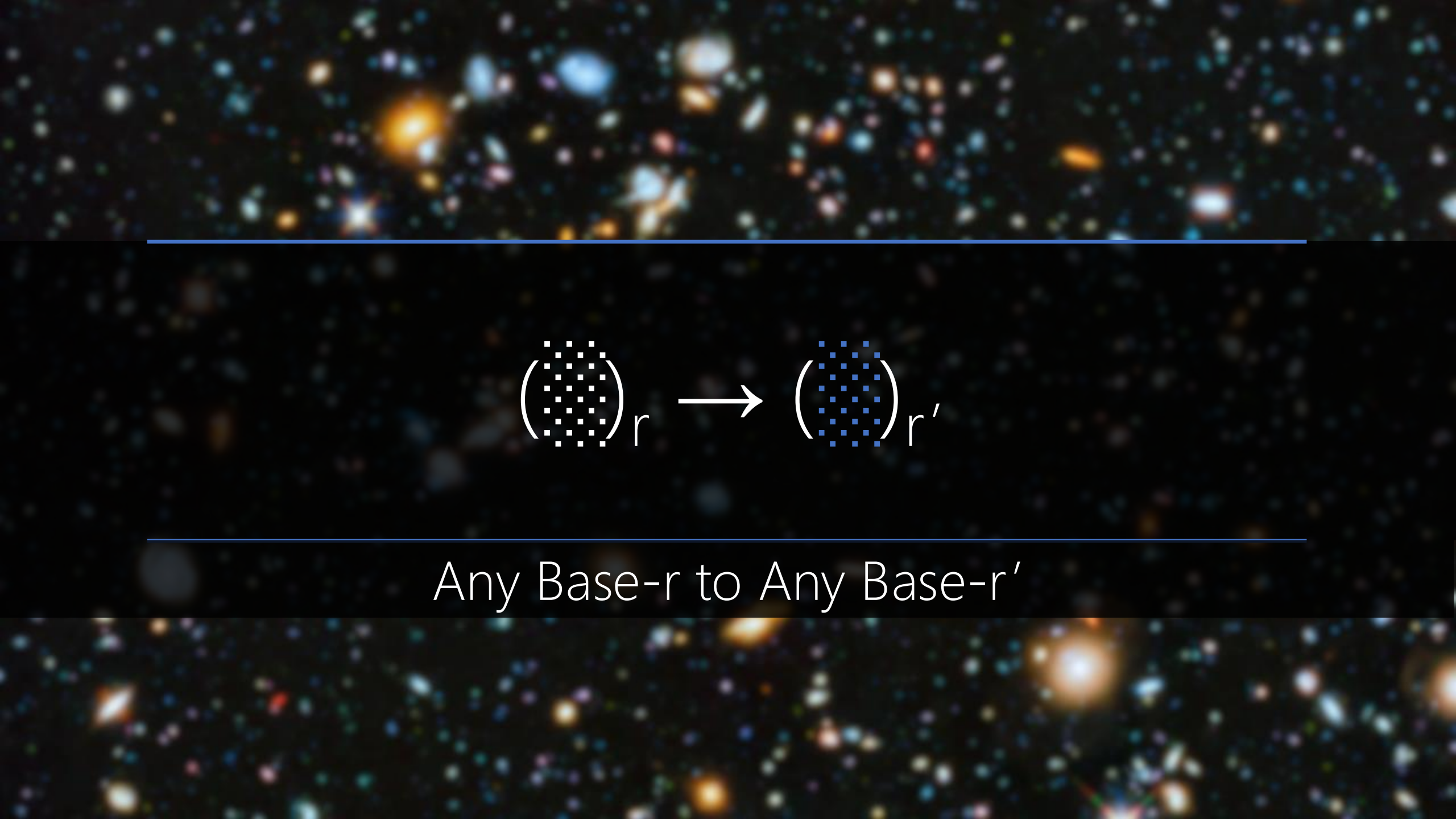
(111001110010111110110011)₂

Less space the better!
But why binary took the stage?

$$\left(\begin{array}{c} \text{5x5 grid of white squares} \end{array} \right)_r \leftarrow \left(\begin{array}{c} \text{10x10 grid of blue squares} \end{array} \right)_{10}$$

1. Divide quotients by r
2. Keep the remainders as the digits in r

For $r > 10$, substitute the digits with valid digits, e.g., in Base-16: $15 \rightarrow F$



$$\left(\begin{array}{c} \cdot \\ \cdot \\ \cdot \\ \cdot \\ \cdot \\ \cdot \\ \cdot \\ \cdot \\ \cdot \\ \cdot \\ \cdot \\ \cdot \\ \cdot \\ \cdot \\ \cdot \end{array} \right)_r \longrightarrow \left(\begin{array}{c} \cdot \\ \cdot \\ \cdot \\ \cdot \\ \cdot \\ \cdot \\ \cdot \\ \cdot \\ \cdot \\ \cdot \\ \cdot \\ \cdot \\ \cdot \\ \cdot \\ \cdot \end{array} \right)_{r'}$$

Any Base- r to Any Base- r'



Mediate through Base-10
Base- $r \rightarrow$ Base-10 $\rightarrow$ Base- r'

$$\begin{aligned}
 (1CE5FB3)_{16} &\rightarrow 1 \times 16^6 + C \times 16^5 + E \times 16^4 + 5 \times 16^3 + F \times 16^2 + B \times 16^1 + 3 \times 16^0 \\
 &\rightarrow 1 \times 16^6 + 12 \times 16^5 + 14 \times 16^4 + 5 \times 16^3 + 15 \times 16^2 + 11 \times 16^1 + 3 \times 16^0 \\
 &\rightarrow (30,302,131)_{10}
 \end{aligned}$$

Quotient	Remainder
$30,302,131 \div 8$	3
$3787766 \div 8$	6
$473470 \div 8$	6
$59183 \div 8$	7
$7397 \div 8$	5
$924 \div 8$	4
$115 \div 8$	3
$14 \div 8$	6
$1 \div 8$	1
0	



$$\rightarrow (163457663)_8$$

A deep-field astronomical image showing a vast field of galaxies. The galaxies are of various shapes and sizes, including spiral, elliptical, and irregular forms. They are colored in a variety of hues, including blue, orange, yellow, and white, set against a dark, star-filled background. Two thin, horizontal blue lines are positioned above and below the central text.

FRACTION CONVERSION

$$(0.\overset{\text{white}}{\underset{\text{white}}{\text{dotted}}})_{10} \leftarrow (0.\overset{\text{blue}}{\underset{\text{blue}}{\text{dotted}}})_{10}$$

To Come up with an Algorithm!

What is the digit in the position with significance of 10^{-i} ?

$$(0.26501)_{10} \rightarrow 10^{-1} \rightarrow 2$$

$$(0.26501)_{10} \rightarrow 10^{-2} \rightarrow 6$$

$$(0.26501)_{10} \rightarrow 10^{-3} \rightarrow 5$$

$$(0.26501)_{10} \rightarrow 10^{-4} \rightarrow 0$$

$$(0.26501)_{10} \rightarrow 10^{-5} \rightarrow 1$$

What is the digit in the position with significance of 10^{-i} ?

$$(0.26501)_{10} \rightarrow 10^{-1} \rightarrow 2$$

$$(0.6501)_{10} \rightarrow 10^{-1} \rightarrow 6$$

$$(0.501)_{10} \rightarrow 10^{-1} \rightarrow 5$$

$$(0.01)_{10} \rightarrow 10^{-1} \rightarrow 0$$

$$(0.1)_{10} \rightarrow 10^{-1} \rightarrow 1$$

Divide by 10^{-1} = Multiply by 10?

(0 . 2 6 5 0 1)₁₀

$$0.26501 \div 10^{-1} = 0.26501 \times 10 = 2.6501$$

Divide by 10^{-1} = Multiply by 10?

(0 . 2 6 5 0 1)₁₀

$$0.6501 \div 10^{-1} = 0.6501 \times 10 = 6.501$$

Fraction	Result	Integer Part
0.26501×10	2.6501	2
0.6501×10	6.501	6
0.501×10	5.01	5
0.01×10	0.1	0
0.1×10	1.0	1
0		

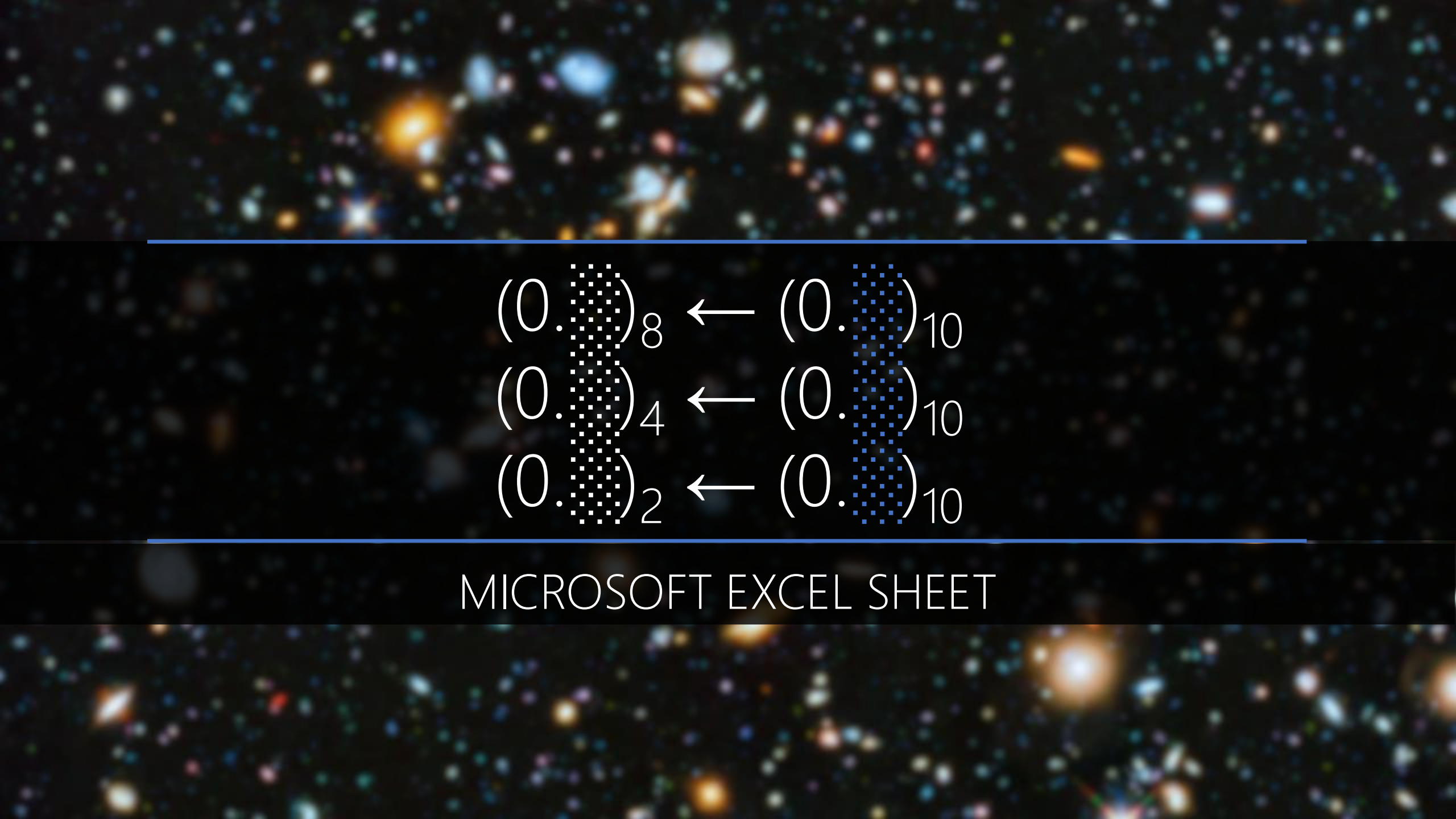


$$(0.\overset{\cdot}{\underset{\cdot}{\text{.}}}\overset{\cdot}{\underset{\cdot}{\text{.}}})_{16} \leftarrow (0.\overset{\cdot}{\underset{\cdot}{\text{.}}}\overset{\cdot}{\underset{\cdot}{\text{.}}})_{10}$$

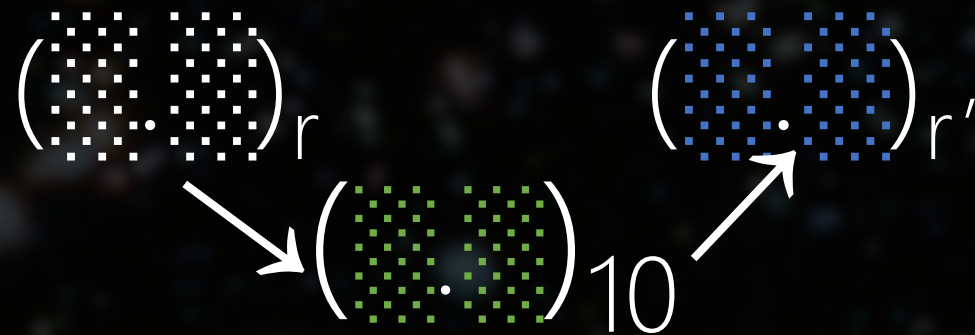
Fraction	Result	Integer Part
0.26501×16	4.24016	4
0.24016×16	3.84256	3
0.84256×16	13.48096	13 = D
0.48096×16	7.69536	7
0.69536×16	11.12576	11 = B
0.12576×16	2.01216	2
0.01216×16	0.19456	0
0.194560006×16	3.11296	3
0.1129601×16	1.807362	1
0.807361603×16	12.91779	12 = C
0.917785645×16	14.68457	14 = E
0.684570313×16	10.95313	10 = A
0.953125×16	15.25	15 = F
0.25×16	4	4
0	0	0



$$(0.26501)_{10} \rightarrow (0.43D7B2031CEAF40)_{16}$$


$$\begin{array}{l} (0.\text{[white dotted box]})_8 \leftarrow (0.\text{[blue dotted box]})_{10} \\ (0.\text{[white dotted box]})_4 \leftarrow (0.\text{[blue dotted box]})_{10} \\ (0.\text{[white dotted box]})_2 \leftarrow (0.\text{[blue dotted box]})_{10} \end{array}$$

MICROSOFT EXCEL SHEET



Integer Part independent of Fraction Part!



Integer Part



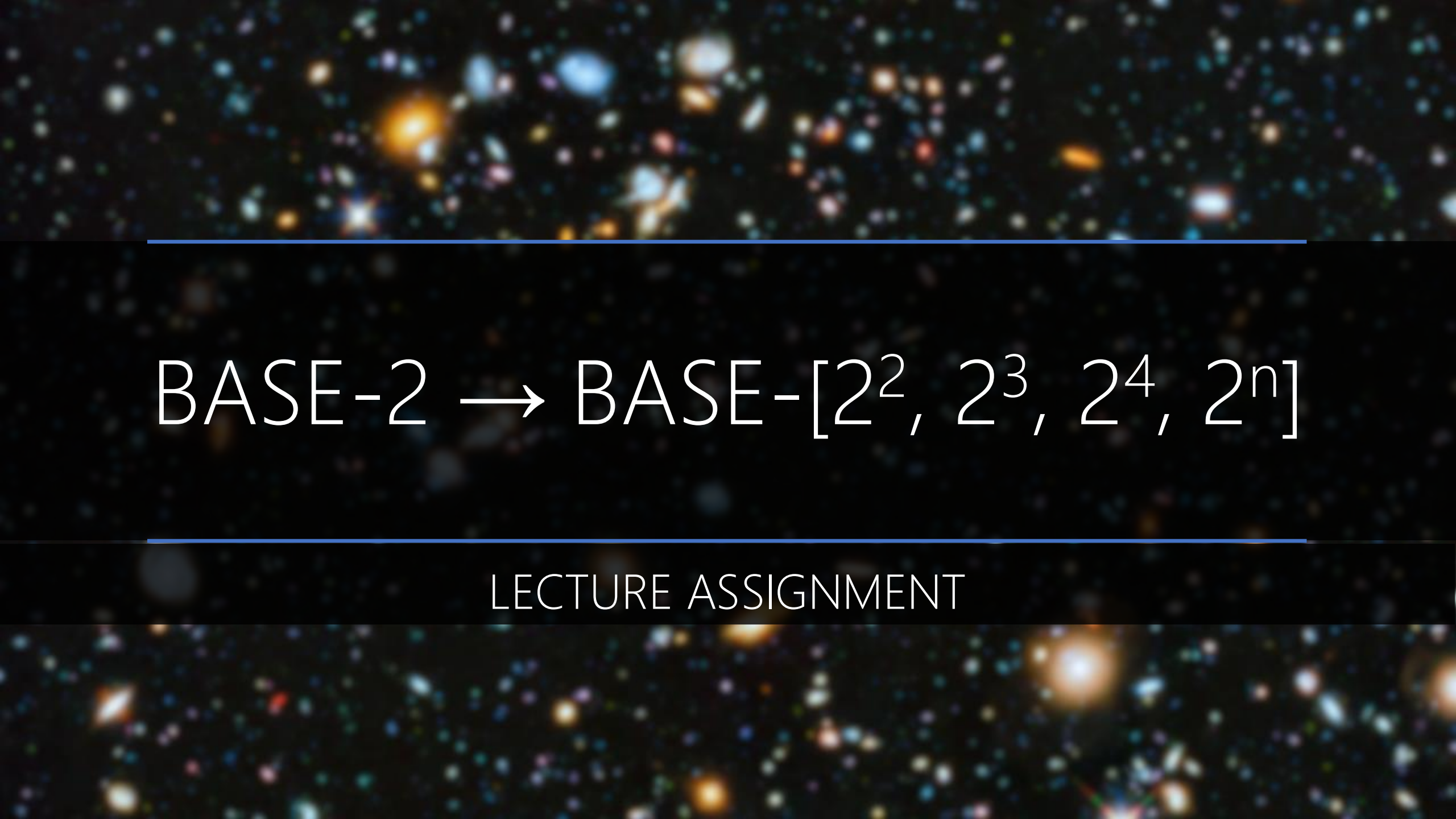
Fraction Part

$(30,302,131.26501)_{10}$

$(163457663.20753....)_8$

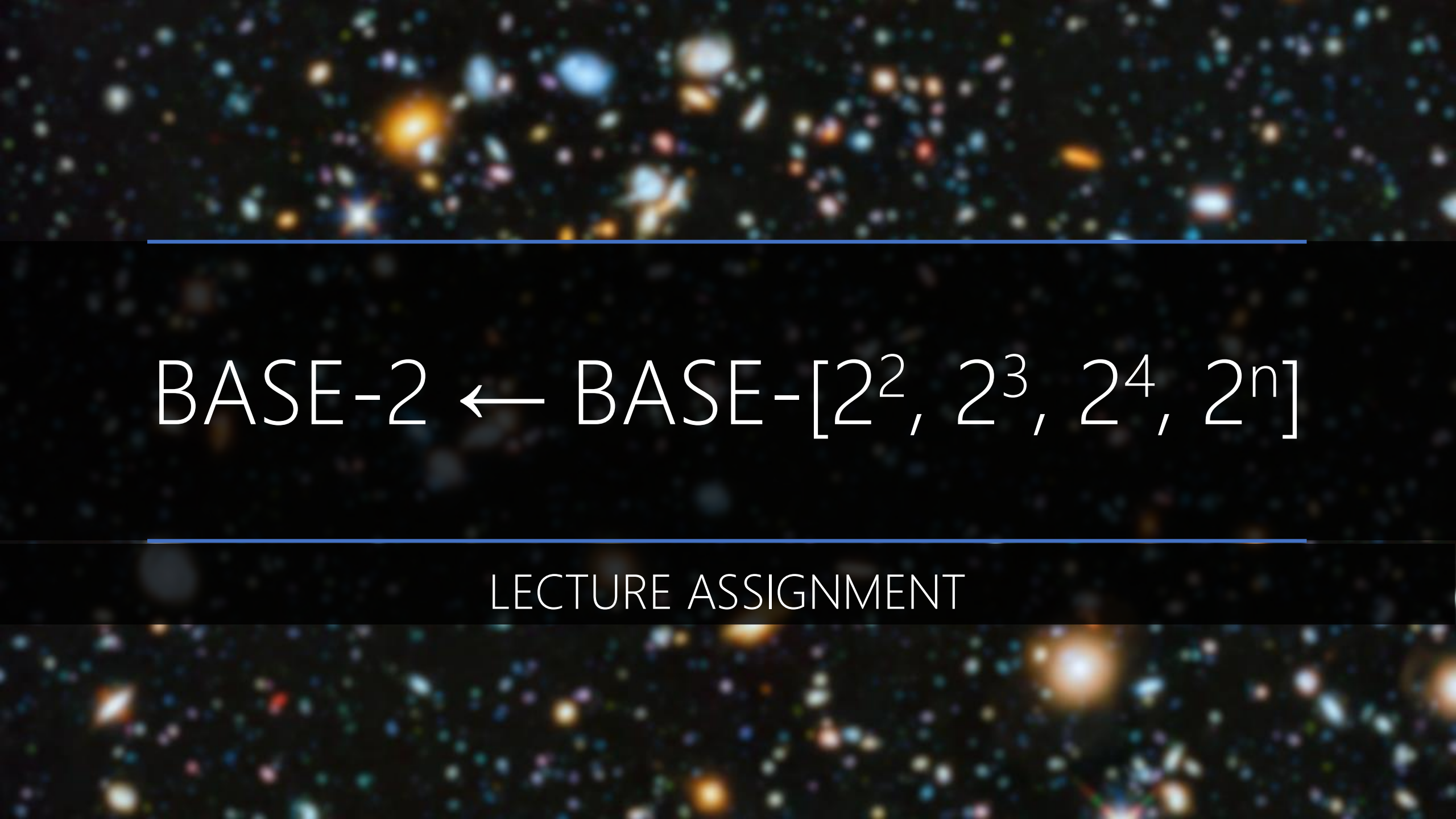
Quotient	Remainder
$30,302,131 \div 8$	3
$3787766 \div 8$	6
$473470 \div 8$	6
$59183 \div 8$	7
$7397 \div 8$	5
$924 \div 8$	4
$115 \div 8$	3
$14 \div 8$	6
$1 \div 8$	1
0	

Fraction	Result	Integer Part
0.26501×8	2.12008	2
0.12008×8	0.96064	0
0.96064×8	7.68512	7
0.68512×8	5.48096	5
0.48096×8	3.84768	3
0.84768×8	6.78144	6
0.78144×8	6.25152	6
0.25152×8	2.01216	2
0.01216×8	0.09728	0
...	...	...

A cosmic background image featuring a dense field of galaxies in various colors (blue, orange, white) against a black space. Two horizontal blue lines are positioned above and below the main text.

BASE-2 $\rightarrow$ BASE-[$2^2, 2^3, 2^4, 2^n$]

LECTURE ASSIGNMENT

A cosmic background image featuring a dense field of galaxies in various colors (blue, orange, white) against a black space. Two horizontal blue lines are positioned above and below the main text.

BASE-2 $\leftarrow$ BASE-[$2^2, 2^3, 2^4, 2^n$]

LECTURE ASSIGNMENT

A cosmic background image featuring a dense field of galaxies in various colors (blue, orange, white) against a black space. A horizontal blue line is positioned above the URL, and another is positioned below it.

<https://planetcalc.com/862/>